

Indirect scalar inferences in negative focus constructions

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Abstract

Indirect scalar inferences are generally assumed to be derived by the same mechanism of implicature calculation that generates direct scalar inferences, whether this mechanism is pragmatic or grammatical in nature. However, this study reveals a critical distinction within indirect scalar inferences, identifying two distinct types: (i) those resembling direct scalar inferences in being defeasible or absent in certain embedded environments; and (ii) those that do not exhibit the characteristic context dependence of implicatures. I propose that the latter type of indirect scalar inferences should be considered a phenomenon *sui generis*. These inferences do not arise from implicature calculation but rather stem from an existential presupposition triggered by certain negative focus constructions. Because of their presuppositional nature, they persist even in downward-entailing environments and in echo denial. To capture the relation between the type of negative construction and the indirect scalar inference it gives rise to, this paper introduces a classification of negative constructions based on their focus sensitivity. It is argued that presuppositional indirect scalar inferences arise only in constructions that conventionally require association with narrow focus. This generalization is supported by cross-linguistic evidence from nine languages that employ a wide range of negative constructions.

Keywords scalar inference, presupposition, implicature, negation, focus, cross-linguistic semantics

1. Introduction

Scalar inferences can be classified into two types: direct and indirect. A direct scalar inference (DSI) typically arises when an assertion containing a weak scalar item is asymmetrically entailed by an alternative statement with the corresponding strong scalar item. For instance, within the scale of ⟨some, all⟩, ‘some’ is the weaker item. In example (1), the assertion with ‘some’ (1-a) is logically weaker than the corresponding alternative with ‘all’ (1-b). Consequently, asserting (1-a) often implies that the speaker does not believe (1-b) to be true.

- (1) a. John ate some of the cookies.
 ↪ John did not eat all of the cookies.
 b. Relevant alternative: John ate all of the cookies.

An indirect scalar inference (ISI), by contrast, typically arises when an assertion containing a strong scalar item is asymmetrically entailed by an alternative statement with the corresponding weak scalar item. This happens when the scalar item occurs in a downward-entailing (DE) environment such as the scope of negation, where entailment relations are reversed. In example (2), the assertion with 'all' (2-a) is logically weaker than the alternative with the existential quantifier (2-b), which is realized here as *any* because of the polarity sensitivity of *some*. As a result, asserting (2-a) often implies that the speaker does not believe (2-b) to be true.

- (2) a. John didn't eat all of the cookies.
 ↪ John ate some of the cookies.
 b. Relevant alternative: John didn't eat any of the cookies.

According to the pragmatic approach, scalar inferences arise from reasoning about why the speaker chose a weaker utterance over a stronger relevant alternative (Horn, 1972; Gazdar, 1979, among others). On this approach, the ISI in (2-a) arises because the listener reasons that the speaker could have uttered the stronger alternative in (2-b). Given that the speaker instead opted for the weaker statement in (2-a), the listener infers that the speaker either does not believe, or lacks sufficient evidence to assert, the stronger statement in (2-b) (Levinson, 2000; Russell, 2006). By contrast, on the grammatical approach, scalar inferences are derived by covert exhaustification within the grammar (Chierchia, 2004; Chierchia et al., 2012, among others). In (2-a), the ISI is derived by inserting an exhaustification operator above negation, thereby excluding the stronger alternative in (2-b) (Chierchia, 2004; Magri, 2011; Romoli, 2012). Thus, despite theoretical differences, both approaches derive ISIs in a manner analogous to DSIs.¹

In this paper, I argue that there are, in fact, two distinct types of ISIs. On the one hand, certain ISIs pattern with DSIs and are presumably derived through the same underlying mechanism, whether pragmatic or grammatical in nature. On the other hand, there exists a separate class of ISIs that does not exhibit the typical context dependence associated with implicatures. I contend that this second type of ISI constitutes a fundamentally different phenomenon, generated through an alternative pragmatic mechanism distinct from that responsible for other scalar inferences. This study specifically examines ISIs where 'all' occurs below negation (i.e., 'not all'), as in (2-a). However, the analysis presented here can potentially extend to ISIs arising from other scalar expressions such as 'not many'.

Consider the contrast between (3-a) and (3-b). The preceding context establishes that no student failed. In this setting, example (3-a) is acceptable, since it does not imply that any of the students failed (no ISI). By contrast, example (3-b) strongly suggests that some

¹ Experimental studies on the processing of ISIs provide conflicting evidence as to whether ISIs are processed the same way as DSIs. Studies consistently show that DSIs incur a processing cost: in truth-value judgment tasks, participants take significantly longer to respond based on strengthened meanings (involving DSIs) than on literal meanings (Bott and Noveck, 2004; Bott et al., 2012). Conversely, Romoli and Schwarz (2015) and Marty et al. (2020) report the opposite pattern for ISIs, where responses based on strengthened meanings (involving ISIs) were faster than those based on literal meanings. Cremers and Chemla (2014) initially found similar faster responses for ISIs (Experiment 1), but they argue these results stemmed from experimental confounds. After introducing appropriate controls (Experiment 2), ISIs displayed processing patterns more akin to DSIs.

students did fail (an ISI), thereby contradicting the common ground. This explains why only (3-a) is felicitous here.²

- (3) *Context: Mary teaches Intro to Statistics. She accidentally gave the students the exam for Advanced Statistics. She expected that they would all fail, but surprisingly, they all passed.*

Mary (to her students): You should all be proud of yourselves for passing the exam. Frankly, ...

- a. I was surprised that you didn't all fail!
- b. #I was surprised that not all of you failed!

Notice that the embedded clauses in (3-a) and (3-b) have the same plain meaning but differ in their syntactic structure: the former involves a floating quantifier construction, whereas the latter employs a negated quantifier phrase (QP) construction. When unembedded, both constructions give rise to the same ISI, as demonstrated in (4). However, the contrast in (3) indicates that these ISIs are not entirely analogous.

- (4) a. You didn't all fail.
 ↪ Some of you failed.
 b. Not all of you failed.
 ↪ Some of you failed.

Consider another case that illustrates the two types of ISIs. In (5), the context establishes that Bob invested none of his savings in the stock recommended by John. In (5-a), where the QP appears in object position, no ISI arises that Bob invested some of his savings in that stock. By contrast, example (5-b) does give rise to such an inference, rendering it infelicitous in this context.

- (5) *Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value plummets to zero.*

Bob says:

- a. Thank goodness I didn't invest all of my savings in that stock!
- b. #Thank goodness not all of my savings were invested in that stock!

² The verb *surprised* is a factive Strawson-Downward-Entailing (SDE) predicate (von Stechow, 1999). Embedded scalar inferences within factive SDE contexts typically exhibit an intriguing pattern: the scalar inference influences the presupposition but not the assertion (Russell, 2006; Simons, 2006; Gajewski and Sharvit, 2012; Spector and Sudo, 2017). In (3-b), the presupposition is that not all, but some, of the students failed, whereas the assertion remains based on the literal meaning. If 'not all' were interpreted as 'not all (but some)' at the assertion level, the sentence would convey that Mary expected either none or all of the students to fail, which is not a salient interpretation of (3-b). See Subsection 5.3.2 for further discussion of scalar inferences in factive SDE contexts.

Additionally, one might consider whether the contrast between (3-a) and (3-b) could result from differences in scope relations rather than from the presence of an ISI. Specifically, it might be argued that the quantifier takes wide scope over negation [$\forall > \neg$] in (3-a), while it takes narrow scope [$\neg > \forall$] in (3-b). However, even if inverse scope were possible in (3-a), which is not certain, such a scope configuration does not accurately capture the actual interpretation. This can be illustrated by attempting a paraphrase using *none*, which is equivalent to a reading where the quantifier takes wide scope [$\forall > \neg$]: *I was surprised that none of you failed!* is not an accurate paraphrase of (3-a). The reason is that (3-a) conveys that Mary expected all of the students to fail, whereas the paraphrase with *none* only implies that she expected some students would fail. The same diagnostic can be applied to other examples presented in this paper, confirming that in all cases negation indeed takes wide scope over the quantifier, thus reinforcing the conclusion that the contrasts observed are genuinely attributable to the presence or absence of an ISI.

I argue that the robustness of an ISI depends on the type of negative construction in which the scalar item occurs—specifically, its focus sensitivity. According to Beaver and Clark (2003, 2008), focus-sensitive expressions can be divided into two major classes: (i) those whose focus sensitivity is conventionalized and lexically encoded (e.g., *only* and *even*); and (ii) those that exhibit non-conventionalized focus sensitivity (e.g., *always* and counterfactual conditionals).³ Standard negation, the basic construction a language employs to negate verbal clauses (Miestamo, 2005), is generally considered to be non-conventionally focus-sensitive (see Fălăuș 2020 for a review).

In this paper, I propose a new distinction within the domain of negative constructions. While standard negation is indeed non-conventionally focus-sensitive, certain negative constructions (often complex ones) are in fact conventionally focus-sensitive. These constructions require association with narrow focus, meaning that focus must fall on an element smaller than a full clause. I refer to such constructions as negative focus constructions (NFCs). By contrast, negative constructions that do not require narrow focus are referred to here as non-NFCs.

As an illustration, consider the difference between (6-a) and (6-b). Example (6-a) involves a standard negation construction, which is not an NFC. This construction does not require focus on a subclausal element like *return* or *change*. On its most natural reading, (6-a) merely denies the claim that the vending machine returns change. In contrast, the negated cleft in (6-b) is an NFC, and it must associate with narrow focus—specifically, with the cleft phrase (the pivot). Moreover, unlike a non-NFC, an NFC triggers an existential presupposition, as exemplified by (6-b), which presupposes that something returns change.

- (6) a. The vending machine doesn't return change.
 b. It's not the [vending machine]_F that returns change.

Arguably, negated clefts should not be classified as a distinct type of negative construction, as they seem to be nothing more than regular clefts embedded under sentential negation. However, other languages have negative constructions that lack positive counterparts, and yet they evoke the same inferences as negated clefts (see Subsection 2.1.1). Hence, the categories of NFCs and non-NFCs capture an important distinction within the typology of negative constructions.

I argue that the contrasts in (3) and (5) are related to whether the universal quantifier occurs in an NFC or a non-NFC. The standard negation construction in the (a) examples (e.g., *You didn't all fail*) is not an NFC, whereas the negated QP construction in the (b) examples (e.g., *Not all of you failed*) is considered an NFC (see Subsection 2.3). I propose the empirical generalization in (7).

(7) *Empirical generalization*

In sentences where 'all' occurs in the scope of negation (i.e., 'not all'), the robustness of the ISI 'not all (but some)' depends on the type of negative construction:

- If the negative construction is a non-NFC, the ISI is suspendable and context-dependent.
- If the negative construction is an NFC, the ISI is obligatory and not affected by contextual considerations.

³ Beaver and Clark (2008) further distinguish two types of non-conventional focus sensitivity: quasi association with focus and free association with focus. Negation and other propositional operators belong to the former category, whereas quantificational adverbs like *always* belong to the latter. This paper is only concerned with the opposition between conventionalized and non-conventionalized focus sensitivity, so this finer distinction will not play a role in the analysis.

The robustness of an ISI depends on the type of negative construction because the inference arises differently in each case. In non-NFCs, such as (3-a) and (5-a), the ISI is derived by implicature calculation, and therefore its presence is contingent upon contextual factors. By contrast, in NFCs, such as (3-b) and (5-b), the ISI arises through an independent route that does not rely on implicature calculation. Specifically, NFCs trigger an existential presupposition, and it is from this presupposition that the ISI emerges. For example, just like *It's not [Bob's]_F savings that were invested in that stock* presupposes that someone's savings were invested in the stock, *Not [all]_F of Bob's savings were invested in that stock* presupposes that some of Bob's savings were invested in the stock. Because this ISI stems from an existential presupposition rather than implicature calculation, it persists even in contexts where other scalar inferences may be suspended.⁴

This paper is structured as follows. Section 2 develops the empirical generalization in (7). It introduces a typology of negative constructions based on their focus sensitivity (Subsection 2.1), relates this typology to the robustness of ISIs (Subsection 2.2), and supports the proposal with additional English examples (Subsection 2.3). Section 3 tests whether the generalization extends beyond English, drawing on cross-linguistic data from nine languages.

Building on this empirical foundation, Section 4 presents a theoretical account of robust ISIs. It explains why existential inferences behave differently in NFCs and non-NFCs (Subsection 4.1) and shows how this distinction accounts for the greater robustness of ISIs triggered by NFCs (Subsection 4.2).

Section 5 addresses three remaining issues: (i) cases where ISIs triggered by NFCs appear to be cancelable, challenging the empirical generalization (Subsection 5.1); (ii) the classification of negated QP constructions as NFCs, even though this is difficult to establish using the diagnostics introduced earlier (Subsection 5.2); and (iii) the relation between robust ISIs and other cases of obligatory scalar inferences, including Magri effects and scalar inferences in factive SDE environments (Subsection 5.3). Finally, Section 6 concludes and outlines directions for future research. An appendix provides additional details on the cross-linguistic data discussed in Section 3.

2. Empirical generalization

2.1. Focus sensitivity of negative constructions

A central goal of this paper is to test the empirical generalization in (7). To do so, it is essential to establish independent criteria for classifying negative constructions as NFCs or non-NFCs. This section develops a typology of negative constructions for this purpose, ensuring that the classification is based on syntactic and semantic properties rather than on scalar inferences.

⁴ For the purpose of this paper, I adopt the standard view that ordinary scalar inferences are not presuppositions, while remaining agnostic as to their pragmatic status—i.e., conversational implicatures in the Neo-Gricean approach (Horn, 1972; Gazdar, 1979, among others) or part of the asserted meaning in most versions of the grammatical approach (Chierchia, 2004; Chierchia et al., 2012, among others).

However, Bassi et al. (2021) propose that ordinary scalar inferences are, in fact, presuppositions triggered by a presuppositional exhaustification operator. In principle, the analysis proposed in this paper is compatible with this view. Even under the assumption that both types of ISIs are presuppositions, they are nonetheless derived in different ways: in NFCs, the ISI emerges from an existential presupposition, while in non-NFCs, it is triggered by an exhaustification operator. See Subsection 4.2.1 for further discussion.

2.1.1. NFCs

NFCs are negative constructions that obligatorily associate with narrow focus, meaning they require a focused element within their scope that is smaller than a full clause. In English, the primary NFC is a negated cleft. Like its positive counterpart, a negated cleft requires that the cleft phrase, or an element within it, be in focus. Furthermore, negated clefts, just like positive clefts, evoke focus alternatives and trigger an existential presupposition (Horn, 1981; Dryer, 1996; Rooth, 1999), as illustrated in (8).

- (8) a. It wasn't [my]_F dog that bit the mailman.
 ↗ Someone's dog bit the mailman.
 b. It wasn't the [mailman]_F that my dog bit.
 ↗ My dog bit someone.

Another type of NFC involves constituent negation of the focused element.⁵ In Hebrew, standard negation is expressed by placing the negative particle *lo* before the finite verb, as shown in (9-a). In contrast, constituent negation occurs when *lo* directly modifies the focused constituent, while the entire negated constituent tends to appear clause-initially, as seen in (9-b). Like negated clefts, the construction in (9-b) qualifies as an NFC because it obligatorily associates with narrow focus. However, unlike negated clefts, Hebrew constituent negation does not have a corresponding positive counterpart.

- (9) Hebrew
 a. ha-kelev šel-i lo našax et ha-davar.
 the-dog POSS-1SG NEG bit ACC the-mailman
 'My dog didn't bite the mailman.'
 b. lo ha-kelev [šel-i]_F našax et ha-davar.
 NEG the-dog POSS-1SG bit ACC the-mailman
 'It wasn't my dog that bit the mailman.'

Another variety of NFC encompasses negative constructions that incorporate a focus marker. Korean has two distinct standard negation constructions: the short-form negation employs a pre-verbal negative particle, while in the long-form negation, the negative prefix *ani-* attaches to the auxiliary verb *ha* 'do', and the lexical verb is followed by the nominalizing suffix *-ci*, known as the suspensive form in traditional grammar (Miestamo, 2005, 17). The long-form standard negation construction is demonstrated in (10-a). In a superficially similar construction, the contrastive focus marker *-nun* is affixed to *-ci*, as in (10-b). This construction (sometimes termed *V-ci-nun*) forces wide scope of negation over preceding scope-bearing elements (Sells and Kim, 2006; Park and Dubinsky, 2020).

- (10) Korean (Eunsun Jou, p.c.)
 a. na-uy kay-ka cipaywon-ul mwul-ci anh-ass-ta.
 1SG-POSS dog-NOM mailman-ACC bite-NMLZ NEG-PST-DECL
 'My dog didn't bite the mailman.'

⁵ The term "constituent negation" has been used in different ways in the literature. It often refers to the distinction between sentential negation and constituent negation proposed in Klima (1964), which is in fact concerned with scope relations rather than syntactic constituency (Zeijlstra, 2022a, 5-6). In this paper, I employ the term "constituent negation" in a syntactic sense to describe constructions where the negative marker modifies the focused constituent rather than appearing in its usual position in standard negation.

- b. [na-uy]_F kay-ka cipaywon-ul mwul-ci-nun anh-ass-ta.
 1SG-POSS dog-NOM mailman-ACC bite-NMLZ-CNTR NEG-PST-DECL
 'It wasn't my dog that bit the mailman.'

According to Sohn (1995, 22), the contrastive focus marker *-nun* can associate with any focused element within the clause, as seen in (11). Crucially, it cannot associate with broad focus over the entire clause. Consequently, this construction is considered an NFC. Furthermore, similarly to constituent negation in Hebrew, this NFC does not have a positive counterpart. Specifically, when *-nun* is attached to the verb in a positive sentence, it cannot associate with a non-adjacent focused element in the manner observed in (10-b) and (11) (Sohn, 1995, 70).⁶

- (11) Korean (Sohn, 1995, 22)
 John-i ecey chayk-ul sa-ci-nun anh-ass-ta.
 John-NOM yesterday book-ACC buy-NMLZ-CNTR NEG-PST-DECL
 Reading 1: 'It is not John who bought a book yesterday.'
 Reading 2: 'It is not yesterday that John bought a book.'
 Reading 3: 'It is not a book that John bought yesterday.'
 Reading 4: 'It is not the act of buying that John did yesterday with a book.'

Yet another diagnostic for identifying NFCs is that they trigger a potential exhaustivity inference. According to Amiraz (2025a), negated clefts imply that the rejected alternative was under consideration as a possible exhaustive answer to the question addressed by the cleft. When this condition is not met, negated clefts become infelicitous. This is illustrated in (12), where the sentence implies that Mary could have been the only guest at the party—a claim that contradicts the established common ground.

- (12) *Context: Jane invited ten of her friends to a party last night. Usually, almost everyone shows up to Jane's parties. John missed the party this time but believes Bill attended.*
 John: Who was at the party last night?
 Bill: Actually, I wasn't there either. #But I do know that it wasn't [Mary]_F (who was at the party). She told me earlier that she had the flu. (Amiraz, 2025a)

Standard negation, by contrast, does not give rise to a potential exhaustivity inference. As a result, (13) is felicitous in the same context (especially if Bill has some reason to believe that John is particularly interested in whether Mary attended the party they missed).

- (13) Bill: Actually, I wasn't there either. But I do know that Mary wasn't at the party. She told me earlier that she had the flu.

This constraint also applies to other NFCs, as illustrated by (14) from Hebrew, which features constituent negation.

⁶ The contrastive marker *-nun* can attach directly to the focused constituent, but then it does not have an exhaustive meaning like a cleft but rather implies ignorance, as demonstrated in (i).

- (i) Korean (Kim, 2018, 469)
 Q: Did all of your friends pass the test?
 A: [Mary]_F-nun tongkwa-hay-ss-eyo.
 PN-CNTR pass-do-PST-DECL
 'Mary passed (perhaps others did not).'

- (14) Hebrew
 #lo meri hayta b-a-mesiba.
 NEG Mary be.PST.3SG.F at-the-party
 'It wasn't Mary who was at the party.'

2.1.2. Non-NFCs

The class of non-NFCs encompasses various negative constructions that do not require the presence of a focused element smaller than a full clause. As the name suggests, these constructions are defined by what they are not, and beyond that, they do not have much in common. The prime example of a non-NFC is standard negation. For instance, (15) can be used to deny the claim that the speaker's dog bit the mailman without evoking alternatives such as those raised by a negated cleft in (8). Another example of a non-NFC is *It is not the case that p*, exemplified in (16), which does not require association with narrow focus either. Other types of non-NFCs include, among others, clauses with negative quantifiers like *no* (17) and negative adverbs such as Hebrew *terem* 'not yet' (18), as well as negative existential constructions (19).

- (15) My dog didn't bite the mailman. (standard negation)
 (16) It is not the case that my dog bit the mailman. (superordinate negation)
 (17) No dog bit the mailman. (negative quantifier)
 (18) Hebrew
 ha-kelev šel-i terem našax et ha-davar.
 the-dog POSS-1SG not.yet bit ACC the-mailman
 'My dog hasn't bitten the mailman yet.' (negative adverb)
 (19) There are no mailman-biting dogs in this neighborhood. (negative existential)

Even though non-NFCs are not conventionally focus-sensitive, most of them do allow a focused element to occur within their scope. Such sentences often generate existential inferences akin to those found in NFCs. For example, the sentences in (20) tend to give rise to similar inferences to those in (8). However, as will be demonstrated in Subsection 4.1, these existential inferences are cancelable implicatures rather than presuppositions, unlike those generated by NFCs.

- (20) a. [My]_F dog didn't bite the mailman.
 ~> Someone's dog bit the mailman.
 b. My dog didn't bite the [mailman]_F.
 ~> My dog bit someone.

2.2. Two types of ISIs

In Subsection 2.1, I introduced a distinction between two classes of negative constructions based on their focus sensitivity: NFCs and non-NFCs. I propose that this distinction correlates with the pragmatic status of the ISI triggered by sentences in which a universal quantifier appears under negation. Table 1 summarizes the predictions: in non-NFCs, ISIs are derived as implicatures, while in NFCs they are presuppositions.

In non-NFCs, the existential inference triggered by a focused DP, as in (21-a), and the ISI triggered by a universal quantifier, as in (21-b), are both implicatures.

- (21) a. My dog didn't bite the [mailman]_F.
 ~> My dog bit someone. (implicature)

Table 1. Pragmatic status of existential inferences and ISIs based on construction type

	Association with focused DP	Association with universal QP
Non-NFC	existential implicature	ISI is an implicature
NFC	existential presupposition	ISI is a presupposition

- b. John didn't eat all the cookies.
 ↪ John ate some of the cookies. (implicature)

In contrast, in NFCs, both types of inferences are presuppositions: the existential inference in (22-a) and the ISI in (22-b) (from the Corpus of Contemporary American English (Davies, 2008)), where a universal QP appears in a negated cleft.

- (22) a. It wasn't the [mailman]_F that my dog bit. (presupposition)
 ↪ My dog bit someone.
 b. Look on the bright side. It's not [every]_F student who has access to their teacher 24 hours a day. [COCA: TV; 2001]
 ↪ Some students have access to their teacher 24 hours a day. (presupposition)

In English, examples of non-NFCs with a universal quantifier under negation include, but are not limited to, the following constructions: floating quantifier constructions (23-a), clauses with a QP in object position (23-b), superordinate negation (23-c), and inverse-scope constructions (23-d).

- (23) a. The cookies did not all burn. (floating quantifier construction)
 b. I didn't burn all the cookies. (QP in object position)
 c. It is not the case that all of the cookies burned. (superordinate negation)
 d. All the cookies didn't burn. (inverse-scope construction)

Constructions such as those in (23-b)–(23-d) are classified as non-NFCs because they involve standard negation, which does not require narrow focus. This can be seen by comparing them to their counterparts without a quantifier. For example, since (24) allows both broad focus, as in (24-a), and narrow focus, as in (24-b), it does not qualify as an NFC. Crucially, under broad focus, the sentence does not imply that the speaker burned something else, confirming that the construction does not obligatorily associate with narrow focus.

- (24) a. I didn't burn the cookies—they are just fine.
 b. I didn't burn the [cookies]_F—I burned the muffins.

Although the floating quantifier construction in (23-a) lacks a non-quantified counterpart, it should still be classified as a non-NFC. This construction also involves standard negation, and floating quantifiers themselves do not independently trigger an existential presupposition, as illustrated in (25). Accordingly, there is no basis for analyzing (23-a) as an NFC.

- (25) Did the cookies all burn?
 ↯ Some of the cookies burned.

NFCs in English include negated clefts (26-a) and negated QP constructions (26-b).

- (26) a. It's not [every]_F student who has access to their teacher 24 hours a day. (negated cleft)
 b. Not [all]_F of the cookies burned. (negated QP construction)

Negated clefts are classified as NFCs because their counterparts with a focused DP clearly exhibit the properties of NFCs, as shown in (8). In contrast, it is much more challenging to demonstrate that negated QP constructions are NFCs. This difficulty arises from the fact that English does not permit a focused DP in the corresponding position, as illustrated in (27).⁷ As a result, evidence for treating these constructions as NFCs must come from other languages. To avoid digression, I postpone this discussion until Subsection 5.2.⁸

- (27) *Not the [cookies]_F burned. (Intended: 'It wasn't the cookies that burned.')

2.3. Additional examples in English

The distinction between ISIs derived through implicature calculation and those arising from an existential presupposition can be difficult to detect in unembedded clauses. As demonstrated in (28) and (29), both types of constructions typically trigger ISIs in such contexts. Indeed, experimental evidence confirms that even non-NFCs like (28) reliably give rise to ISIs (Romoli and Schwarz, 2015).

- (28) Non-NFCs
 a. The cookies did not all burn. (floating quantifier construction)
 ↪ Some of the cookies burned.
 b. I didn't burn all the cookies. (QP in object position)
 ↪ I burned some of the cookies.
- (29) NFCs
 a. Not [all]_F of the cookies burned. (negated QP construction)
 ↪ Some of the cookies burned.
 b. It's not [every]_F cookie that leaves its mark on history. (negated cleft)
 ↪ Some cookies leave their mark on history.

However, the difference between presuppositional ISIs and implicature-based ISIs becomes evident when these constructions appear in contexts where scalar implicatures are typically suspended. In these cases, ISIs triggered by NFCs persist, while those triggered by non-NFCs tend to disappear, revealing their different underlying status.

First, scalar inferences are typically suspended in non-factive SDE environments (Chierchia, 2004; Chierchia et al., 2012), such as antecedents of conditionals. The contrast between non-NFCs and NFCs is particularly striking in conditional warnings, where non-NFCs are acceptable (30-a) (from the Movie Corpus (Davies, 2019)), whereas NFCs tend to be infelicitous (30-b).⁹ It is not entirely clear whether the ISI in (30-b)—namely, the inference that some of the addressees will leave immediately—projects from the antecedent

⁷ See Wu (2025) for an analysis of restrictions on constituent negation in English. Also see Subsection 5.2 for discussion.

⁸ In Subsection 5.2, I argue that in Hebrew, constituent negation of a focused DP has the same semantics as negated clefts—and crucially, the same applies to negated QP constructions. Extending this analysis to English helps explain an otherwise puzzling restriction on focus placement in English negated QP constructions.

⁹ A similar example, though not involving a conditional warning per se, is given below:

- (i) a. And if you don't all stop acting like a bunch of crazy neurotics, I'm going to start acting like one myself. [The Movie Corpus; 1946]

of the conditional or is locally accommodated within it (See Subsection 4.2.2). Nonetheless, this inference appears to be the culprit for the unacceptability of (30-b).¹⁰

- (30) a. If you don't all leave right now, we're gonna have to start making arrests!
[The Movie Corpus; 1996]
b. #If not all of you leave right now, we're gonna have to start making arrests!

In (31), a clause containing 'all' under negation is embedded within the antecedent of a counterfactual conditional.

- (31) *Context: A football team, with only one win needed to secure the championship, had five matches ahead. Tragically, while en route to what seemed like a certain victory, their plane crashed, claiming the lives of all on board.*
a. This team would have won the title if they hadn't all died in that plane crash.
b. #This team would have won the title if not all of them had died in that plane crash.

Both (31-a) and (31-b) consider counterfactual situations where it is not the case that all team members died in a plane crash. Under the literal (unstrengthened) interpretation, the relevant counterfactual situations include both scenarios where the plane crashes but some members survive and scenarios where nobody dies at all, such as scenarios where the crash never happens and the team arrives safely at the match. Crucially, only example (31-a) naturally allows for the scenario where nobody dies (e.g., no crash occurs). Indeed, this is the most salient interpretation of (31-a), presumably because the closest worlds where the team members don't all die are those where the crash never happens. This suggests that no ISI is generated in (31-a). In contrast, (31-b) only considers situations where some team members did perish in the crash. In other words, the antecedent is interpreted as 'if not all, but some, of them had died in that plane crash'.¹¹ This ISI creates an awkward interpretation, as it suggests that if some of the players had survived the crash, they would have gone on to win another match and secure the title, which seems unlikely. Consequently, (31-b) is somewhat odd within the given context.¹²

The second context where scalar inferences are often suspended is echo denial. Echo denial, also known as emphatic denial (Baker, 1970), occurs in cases where a speaker rejects a prior statement by repeating it with an added negative marker.¹³ For example, in (32), B's response does not amount to an admission of having burned some of the cookies.

(32)

- b. #And if not all of you stop acting like a bunch of crazy neurotics, I'm going to start acting like one myself.

¹⁰ Most weak scalar items are acceptable in conditional warnings, as illustrated with disjunction in (i-a) and with the numeral 'one' in (i-b).

- (i) a. If you talk to the police or the press, you'll be in trouble.
b. If one of you talks to the police, you will all be in trouble.

While it is not straightforward to demonstrate that (i-a) and (i-b) do not trigger scalar inferences, the fact that they are natural aligns with the acceptability of non-NFCs in this context (30-a).

¹¹ The ISI in (31-b) is locally accommodated within the antecedent of the counterfactual conditional, as discussed in Subsection 4.2.2.

¹² The same contrast arises in cases where 'not all' appears in the consequent of the counterfactual conditional:

- (i) a. If the plane hadn't crashed, they wouldn't have all been dead.
b. #If the plane hadn't crashed, not all of them would have been dead.

¹³ Echo denial is often associated with metalinguistic negation (Horn, 1989), but it can also express descriptive negation, as demonstrated in (33).

A: You burned all the cookies!

B: I didn't burn all the cookies—I just gave them a nice crust.

A naturally occurring example of this phenomenon is provided in (33), where the fragment *they are not all men* does not give rise to an ISI, as evidenced by the continuation of the sentence.

- (33) "What does the king want from us?"
 ...
 "He wants us to join the musketeers."
 "Sister, they are all men, I do not want to be part of that."
 "**No, Rene, they are not all men, they are all women.**"¹⁴

From a Gricean perspective, the absence of an ISI can be explained by the nature of echo denial, which typically preserves the lexical material of the preceding utterance. In (33), the listener reasons that the speaker's choice of the weaker alternative (*they are not all men*) over the stronger one (*none of them are men*) reflects a desire to maintain the original form, rather than to convey any implicature. As a result, the speaker's choice does not invite the inference that the stronger alternative is false, and no ISI arises.

I observe that this type of interpretation is not available with a negated QP construction, which is an NFC. In the modified version shown in (34), the negated QP obligatorily triggers an ISI—an inference that is contradicted by the continuation of the sentence. In other words, an NFC cannot be used in echo denial without committing the speaker to an existential statement.

- (34) [...]

 "Sister, all of them are men, I do not want to be part of that."

 #"**No, Rene, not all of them are men, all of them are women.**"

As pointed out to me by an anonymous reviewer, suspending an ISI does not require echo denial of an explicit prior utterance. Rather, an ISI can also be suspended when the sentence rejects an assumption that is implicitly introduced by the preceding discourse. This is illustrated in (35), suggested by the reviewer. In this example, the context renders the proposition 'Our friends all died' discourse-salient, even though it is not explicitly stated by an interlocutor.

- (35) *Context: Some of our friends are on a helicopter, and there are rumors that it crashed. We know that if the helicopter crashed, there is no way anyone could have survived.*
 Someone says:
 a. Luckily, the helicopter didn't crash, so our friends didn't all die.
 b. #Luckily, the helicopter didn't crash, so not all of our friends died.

3. Testing the generalization: Cross-linguistic data

This section tests the empirical generalization in (7), which links the type of negative construction to the robustness of the ISI it triggers. To this end, I examine negative

¹⁴ Foley, Muron & Poe, Mina Anne. 2017. *A treasure trove of new Classics*, p. 73. Pittsburgh, PA: Rosedog Books.

constructions that are either absent or rare in English. The data are drawn from nine languages: Finnish, French, Hebrew, Japanese, Korean, Mandarin Chinese, Bosnian-Croatian-Serbian, Swedish, and Turkish. These languages were selected because they exhibit a wide range of syntactic strategies for expressing propositions of the form ‘Not all X are Y’ (see Amiraz 2025b for a cross-linguistic survey of how languages encode this type of meaning).

The results are summarized in Table 2. Examples and descriptive information about the relevant negative constructions in each language are provided in the appendix. Overall, the results support the proposed generalization: NFCs typically differ from non-NFCs in the type of ISIs they generate. However, two constructions—specifically, in Japanese and Mandarin Chinese—appear to challenge the generalization. These cases are addressed briefly below and in greater detail in the appendix.

Table 2. Relation between negative construction class and ISI type across nine languages

Language	Construction	Class	Type of ISI
Finnish	VS word order	non-NFC	implicature
	inverse scope	non-NFC	implicature
French	floating quantifier	non-NFC	implicature
	inverse scope	non-NFC	implicature
Hebrew	negated QP	NFC	presupposition
	VS word order	non-NFC	implicature
	inverse scope	non-NFC	implicature
	<i>wake de wa nai</i>	non-NFC	impl./pres.
Japanese			
Korean	<i>V-ci-nun</i>	NFC	presupposition
Mandarin Chinese	<i>bu shi</i>	non-NFC	impl./pres.
	floating quantifier	non-NFC	implicature
Bosnian-Croatian-Serbian	VS word order	non-NFC	implicature
Swedish	inverse scope	non-NFC	implicature
	clause-initial negation	non-NFC	implicature
	negated cleft	NFC	presupposition
Turkish	inverse scope	non-NFC	implicature

In what follows, I highlight two languages that illustrate the predictive power of the generalization: Hebrew as a straightforward case, and Swedish as a more complex one.

Example (36-a) from Hebrew features standard negation with VS word order (non-NFC). As predicted, this construction does not necessarily trigger an ISI and is felicitous in the given context. In contrast, the negated QP construction in (36-b) gives rise to a mandatory ‘not all (but some)’ ISI because it is an NFC. This leads to infelicity, given that the common ground entails that the cookies were not even baked.

- (36) Hebrew
- Context: *John was baking cookies. He put the cookies in the oven but forgot to set a timer. Two hours later, he realized he had completely forgotten about the cookies. Rushing to the kitchen, he opened the oven, only to discover he had never turned it on.*
- John later tells his friend:

- a. rak be-mazal lo nisrefu l-i kol ha-ugiy-ot.
 only in-luck NEG burn.PST.3PL DAT-1SG all the-cookie-PL
 'It was only through luck that the cookies did not all burn.'
- b. #rak be-mazal lo kol ha-ugiy-ot nisrefu l-i.
 only in-luck NEG all the-cookie-PL burn.PST.3PL DAT-1SG
 'It was only through luck that not all of the cookies burned.'

Example (37) from Swedish might initially seem to contradict the generalization in (7): the negative particle *inte* precedes the QP in what superficially resembles a negated QP construction—that is, an NFC. Yet the sentence can be felicitously uttered in the given context without triggering an obligatory ISI, unlike its English counterpart in (5-b).

(37) Swedish (Alice Bondarenko, p.c.)

Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value plummets to zero.

Bob says:

Tack och lov att inte alla mina besparingar var investerade in
 thank and praise that NEG all.PL my.PL savings be.PST invest.PTCP.PL in
 den där aktien!
 that there stock
 'Thank goodness not all of my savings were invested in that stock!'

The apparent discrepancy is resolved once we recognize that (37) is structurally ambiguous. In one reading, *inte* modifies the QP, yielding a genuine negated QP construction. Under this interpretation, the sentence triggers an obligatory ISI and is therefore infelicitous in the given context. In an alternative reading, however, *inte* takes scope over the entire clause, and in this structure, the sentence does not trigger an ISI—making it pragmatically felicitous.

According to Teleman et al. (1999, 41), the negative particle *inte* can precede the subject in certain embedded clauses. Thus, Bob could just as well have said (38), where *inte* precedes a definite DP rather than a QP. Crucially, this is an instance of clause-initial clausal negation—not constituent negation of *mina besparingar* 'my savings'. This is evidenced by the fact that (38) does not necessarily imply that someone else's savings were invested in that stock, mirroring the interpretation of the English translation. Accordingly, this construction is classified as a non-NFC.¹⁵

(38) Swedish (Alice Bondarenko, p.c.)

Tack och lov att inte mina besparingar var investerade in den
 thank and praise that NEG my.PL savings be.PST invest.PTCP.PL in that
 där aktien!
 there stock
 'Thank goodness my savings were not invested in that stock!'

¹⁵ Additional evidence that this construction does not require narrow focus on the constituent following *inte* comes from the following example, which does not imply that something else ran into the road:

- (i) Swedish
 Tack och lov att inte älgen sprang ut i vägen.
 thank and praise that NEG moose.DEF run.PST out into road.DEF
 'Thank goodness the moose didn't run out into the road.' [web example]

Given that *inte* may freely precede the subject in this context—as demonstrated by (38)—example (37) is structurally ambiguous between an NFC and a non-NFC interpretation. This ambiguity makes the sentence potentially felicitous in the given context, since the non-NFC reading does not trigger an ISI. As such, this example ultimately reinforces, rather than undermines, the generalization in (7).

Finally, unlike the Swedish case, two constructions remain problematic: *wake de wa nai* in Japanese and *bu shi* in Mandarin Chinese. These constructions do not meet the criteria for NFCs, yet they nonetheless yield robust ISIs in certain contexts. I argue that this mixed behavior is tied to the felicity conditions of the broad-focus interpretation. When broad focus is felicitous, the ISI is an implicature and can therefore be suspended. However, in both languages, the broad-focus interpretation is infelicitous in standalone sentences and in certain embedded environments. When broad focus is pragmatically blocked, only a narrow-focus interpretation remains, and in such cases the constructions effectively behave like NFCs, yielding a presuppositional ISI. This pattern motivates a refinement of the generalization in (7). Rather than a categorical split between NFCs and non-NFCs, the crucial factor is whether narrow focus is required by the construction in a given context. A fuller exploration of the Japanese and Mandarin Chinese data is provided in the appendix.

4. Proposal

4.1. Existential inferences

In Section 2, I argued that NFCs trigger existential presuppositions, while non-NFCs only give rise to existential implicatures when they associate with narrow focus. This distinction was shown to underlie the observed difference in the robustness of the resulting ISIs. The goal of this section is to explain why these two types of existential inferences differ.

This section is structured as follows. Subsection 4.1.1 presents the key empirical patterns that distinguish existential inferences triggered by NFCs from those triggered by non-NFCs. Subsection 4.1.2 reviews the theoretical approaches that have been proposed to explain this contrast and argues in favor of the view where the existential presupposition of NFCs is semantically encoded.

Throughout the discussion, I focus on representative constructions from each class. For NFCs, the main example will be negated clefts, such as (8), which have been extensively studied. Other NFCs, such as constituent negation (9-b) and constructions with focus markers (10-b), differ structurally (and possibly also semantically) but they appear to behave similarly with respect to the existential inferences they trigger. For non-NFCs, I concentrate on standard negation constructions with prosodic focus.¹⁶

4.1.1. The empirical landscape

An unembedded cleft of the form *It is x that P* asserts that *x* is *P* and gives rise to two not-at-issue inferences: an existential inference and an exhaustivity inference (Halvorsen, 1978; Horn, 1981), as illustrated in (39).

(39)

¹⁶ As a matter of terminology, I refer to existential inferences triggered by negated clefts as presuppositions, and those triggered by standard negation with prosodic focus as implicatures. Alternatively, this contrast has been framed in terms of hard versus soft presuppositions (Abusch, 2010), where both inferences count as presuppositions but differ in their degree of cancelability. A different perspective is offered by Romoli (2012, 2015), who argues that soft presuppositions are in fact obligatory scalar implicatures, and thus have a distinct pragmatic status. Further complicating the landscape, Bassi et al. (2021) propose that scalar implicatures in general should be analyzed as presuppositions. Despite these theoretical divergences, what matters for present purposes is simply that there is a systematic difference between the two types of inferences: those triggered by NFCs are more robust than those triggered by non-NFCs.

It was [my]_F dog that bit the mailman.

- a. Assertion: My dog bit the mailman.
- b. Existential inference: Someone's dog bit the mailman.
- c. Exhaustivity inference: Nobody else's dog bit the mailman.

The existential inference projects from entailment-canceling environments such as the scope of negation and possibility modals, as evidenced by the infelicity of (40-a) and (40-b), respectively. Based on this behavior, this inference is generally considered a presupposition (Horn, 1981; Dryer, 1996; Rooth, 1999, among others). Indeed, Abusch (2010) identifies clefts as a prime example of a hard presupposition trigger, given that the existential presupposition cannot be easily suspended.

- (40) a. #It wasn't [my]_F dog that bit the mailman, and neither did anyone else's dog.
The mailman is just lying about being bitten.
- b. #It might have been [my]_F dog that bit the mailman, though it is more likely that the mailman is just lying about being bitten.

However, the cancelability of this presupposition remains a matter of controversy. While Abusch (2010) argues that it cannot be suspended—presenting (41) as evidence—others have challenged this view. For instance, Rooth (1999) considers (42) to be a felicitous example in which the existential presupposition appears to be suspended. To account for such cases, Velleman et al. (2012) suggest that the speaker may provisionally accept the presupposition for the sake of argument, without committing to it epistemically. According to Rooth (1999), cancelation is possible in contexts involving hypothetical reasoning, but not when the communicative goal is information update, as in (40).

- (41) I have no idea whether anyone read that letter. #But if it is [John]_F who read it, let's ask him to be discreet about the content. (Abusch, 2010, 40)
- (42) A: Did someone borrow my Badminton racket?
B: I don't know. If it's [John]_F who borrowed it, you can forget about getting it back in one piece. (Rooth, 1999)

Despite these differences, there is broad agreement that prosodically marked focus tends to trigger a weaker existential inference—one that is more easily canceled. As noted in Subsection 2.1.2, standard negation constructions involving narrow focus typically give rise to such inferences, as illustrated in (43).

- (43) a. [My]_F dog didn't bite the mailman.
↪ Someone's dog bit the mailman.
- b. My dog didn't bite the [mailman]_F.
↪ My dog bit someone.

Crucially, while negated clefts (and other NFCs) are considered hard presupposition triggers, prosodically marked focus is not. This distinction is supported by two main pieces of evidence. First, as shown in (44), an existential implicature can be explicitly denied without yielding a contradiction—unlike in the case of NFCs such as negated clefts, where this kind of denial results in infelicity, as seen in (40).

- (44) a. [My]_F dog didn't bite the mailman, and neither did anyone else's dog. The mailman is just lying about being bitten.
- b. [My]_F dog might have bitten the mailman, though it is more likely that the mailman is just lying about being bitten.

Second, when a question explicitly conveys ignorance about the existential claim, a standard negation construction with a focused constituent can still serve as an appropriate response. In contrast, an NFC is infelicitous in this context, as illustrated in (45). This is because an existential presupposition signals that it is taken for granted that the dog bit someone, whereas the speaker's question explicitly indicates that this is not part of the common ground.¹⁷

- (45) A: Who, if anyone, did the dog bite?
 B: I have no idea, but it certainly didn't bite the [mailman]_F.
 B': #I have no idea, but it certainly wasn't the [mailman]_F that it bit.

It is worth noting in passing that the same distinction applies to the exhaustivity inference. First, experimental studies have shown that exhaustivity inferences triggered by prosodic focus in canonical sentences are more easily canceled than those associated with clefts (see Onea, 2019, 410–413, and references therein). Second, negated clefts and other NFCs give rise to a potential exhaustivity inference, whereas standard negation with narrow focus does not (see Subsection 2.1.1). Thus, although the pragmatic status of cleft exhaustivity remains debated—variously analyzed as a presupposition encoded by the cleft (Velleman et al., 2012; Büring and Kriz, 2013) or as a conversational implicature (Horn, 1981; Pollard and Yasavul, 2016)—it is consistently stronger than that of prosodic focus. This reinforces the broader pattern where clefts, as a type of NFC, allow less flexibility in the inferences they trigger.

4.1.2. Theoretical accounts

This subsection reviews theoretical approaches that aim to explain why existential inferences triggered by NFCs, such as negated clefts, are more robust than those triggered by standard negation with prosodic focus. The proposals fall into two main categories.

The first approach, put forward by Abusch (2010), holds that the existential inference of clefts is semantically encoded, whereas the corresponding inference of prosodic focus is pragmatically derived. According to Abusch, the existential inference of prosodic focus arises from a pragmatic constraint requiring the disjunction of the focus alternatives to be true. For example, consider (46). When interpreted as a partial answer to the QUD “Whose dog bit the mailman?”, the sentence evokes the set of alternatives in (46-a). The disjunction of this set yields the existential implicature in (46-b), which, according to Abusch, is cancelable because it is pragmatically derived. By contrast, in clefts such as (39), the existential inference is a true semantic presupposition, making it harder to suspend.

- (46) [My]_F dog didn't bite the mailman.
 a. Alternative set: {*x*'s dog bit the mailman | $x \in D$ }
 b. Existential implicature: Someone's dog bit the mailman.

The second set of approaches links the contrast to how clefts interact with the QUD. Building on Abusch's account, Abrusán (2016) takes the idea one step further by proposing that the existential presupposition of clefts need not be semantically encoded either. Instead, she argues that the same pragmatic mechanism responsible for generating existential inferences in questions and prosodic focus should also apply to clefts. According to Abrusán, the key difference between clefts and prosodic focus is that clefts place stricter constraints on question–answer congruence than prosodic focus does. Specifically, clefts must associate with a QUD that is recoverable from the content of the cleft clause. For example, in (47),

¹⁷ See Dryer (1996, 487–488) for a similar observation regarding the difference between free focus and clefts in positive sentences.

the denotation of the cleft clause is (47-a), and the relevant QUD is the Hamblin set in (47-b).

- (47) It is [John]_F [who read the letter]_Q. (adapted from Abrusán, 2016, 184)
- a. Cleft clause: $\lambda x. \text{read the letter}(x)$
 - b. QUD: $\lambda p. \exists x[p = \lambda w. \text{read the letter}(x)(w)]$

By contrast, prosodic focus allows greater flexibility in question–answer congruence. For instance, while (46) typically gives rise to an existential implicature, the QUD need not correspond directly to the focus alternatives in (46-a). Instead, the relevant QUD may be an implicated question that gives rise to a vacuous implication—for example, “Whose dog, if anyone’s, bit the mailman?”, as in the exchange in (45). This kind of QUD allows the existential inference to be canceled in an appropriate context. Crucially, such flexibility is not available with clefts. In a sentence like (40-a), the QUD must align with the syntactic content of the cleft clause, making it incompatible with a QUD like “Whose dog, if anyone’s, bit the mailman?”

This account is challenged, however, by the fact that the existential presupposition of clefts does not always match the question generated by the cleft clause. This is illustrated in (48), adapted from Velleman et al. (2012). These examples show that the location of focus inside the cleft phrase determines the resulting existential inference. In (48-a), the entire cleft phrase is focused, yielding a presupposition that corresponds closely to the cleft clause. In (48-b) and (48-c), however, only modifiers of the head noun are focused, resulting in more specific existential inferences. These examples suggest that alignment between the cleft clause and the existential presupposition holds only when the entire cleft phrase is in focus.

- (48) a. It wasn’t [John’s eldest daughter]_F who read the letter.
 $\rightsquigarrow$ Someone read the letter.
- b. It wasn’t [John’s]_F eldest daughter who read the letter.
 $\rightsquigarrow$ Someone’s eldest daughter read the letter.
- c. It wasn’t John’s [eldest]_F daughter who read the letter.
 $\rightsquigarrow$ One of John’s daughters read the letter.

Based on this observation, Velleman et al. (2012) argue that the existential presupposition is generated by the focus alternatives themselves, rather than by the cleft clause as such. Velleman et al. (2012) further develop this idea by offering a pragmatic rationale for why the disjunction of the alternative set must be true. They propose a pragmatic principle stating that a cooperative speaker must reject an unanswerable question as soon as possible. A question is considered unanswerable if it contains no true answer. For instance, consider again the negated cleft *It wasn’t my dog that bit the mailman*. This cleft addresses the QUD “Whose dog bit the mailman?”, which contains alternatives of the form ‘*x*’s dog bit the mailman’. Crucially, the proposition ‘nobody’s dog bit the mailman’ is treated not as a direct answer, but as a rejection of the question. If the speaker believed that no one’s dog bit the mailman, it would be uncooperative to employ a negated cleft, since it presupposes that the question is worth asking. Therefore, by employing a negated cleft, the speaker effectively acknowledges the validity of the QUD and, in doing so, introduces a presupposition that at least one of the alternatives in the QUD is true. This results in a presupposition that someone’s dog bit the mailman, which corresponds to the disjunction of the alternatives in the QUD.

While this account offers a compelling explanation for clefts, it reopens the basic question: why do clefts trigger a strong existential presupposition, whereas prosodic focus yields only a

defeasible implicature? If the inference in both cases arises from the set of focus alternatives, what explains the difference in strength?

I take this asymmetry to suggest that NFCs differ from prosodic focus in their semantics, in the spirit of Abusch (2010). Specifically, I assume that NFCs introduce a focus-sensitive operator, along the lines proposed by Velleman et al. (2012). Unlike Velleman et al. (2012), however, I take this operator to encode not only an exhaustivity presupposition, but also an existential presupposition corresponding to the disjunction of the focus alternatives. The denotation of the cleft operator is given in (49), where *ALT* is a salient subset of the set of focus alternatives. The first conjunct in the presupposition is responsible for the existential inference. The second conjunct yields the exhaustivity inference by requiring all alternatives strictly stronger than the prejacent to be false.

$$(49) \quad \llbracket \text{CLEFT} \rrbracket^c(p) = \lambda w : \exists q \in \text{ALT}[q(w)] \wedge \forall r \in \text{ALT}[r \subset p \rightarrow \neg r(w)].p(w)$$

An unembedded cleft asserts the prejacent *p* and presupposes that no stronger alternative is true. This captures the meaning of clefts, as described in (39). A negated cleft, on the other hand, implies that there is a different true alternative, as the prejacent is asserted to be false, though the speaker might not be able to identify it.

This proposal provides a concrete way of deriving the observed asymmetry between the two types of existential inferences. At the same time, the broader argument about presuppositional ISIs does not hinge on the specific implementation adopted here. Regardless of how the existential presupposition is derived, its presence or absence plays a crucial role in determining the robustness of ISIs triggered by negative constructions. As the next subsection will argue, NFCs trigger robust ISIs because they carry non-cancelable existential presuppositions, whereas non-NFCs trigger defeasible ISIs precisely because they are not presupposition triggers.

4.2. Deriving presuppositional ISIs in NFCs

4.2.1. From an existential presupposition to an ISI

I argue that ISIs triggered by NFCs are more robust than those triggered by non-NFCs because they originate from different sources. In NFCs, the ISI stems from an existential presupposition, whereas in non-NFCs, it results from implicature calculation. This distinction helps explain why ISIs triggered by NFCs persist even in environments where scalar implicatures are typically suspended.

To see how ISIs are generally derived, consider the example in (50-a), which features a non-NFC construction in which a strong scalar item ('all') appears in the scope of negation—a DE environment. In this configuration, the assertion containing 'all' (50-a) is logically weaker than the corresponding alternative with the weak scalar item, realized here as *any* (50-b).

- (50) a. John didn't eat all of the cookies.
 $\rightsquigarrow$ John ate some of the cookies.
 b. Relevant alternative: John didn't eat any of the cookies.

In the pragmatic approach (Horn, 1972; Gazdar, 1979, among others), the inference is derived via Gricean reasoning. The speaker chose not to assert the logically stronger alternative in (50-b). Given that a cooperative speaker is expected to be as informative as required, this choice reflects an intentional avoidance of the stronger statement. This, in turn, invites the inference that the speaker either does not believe or lacks sufficient evidence

that John ate none of the cookies. As a result, the listener is likely to infer that John ate some of the cookies.

In the grammatical approach (Chierchia, 2004; Chierchia et al., 2012, among others), a silent exhaustification operator is inserted above negation (Chierchia, 2004; Magri, 2011; Romoli, 2012). This operator asserts the prejacent and negates the relevant alternatives, resulting in an enriched meaning, as shown in (51).

- (51) EXH [not [John ate all of the cookies]] =
John didn't eat all of the cookies, and John ate some of the cookies.

Despite their differences, both approaches share a key assumption: ISIs are derived by the same mechanism of implicature calculation that produces DSIs (cf. Romoli and Schwarz, 2015). More broadly, scalar inferences arise when stronger alternatives are contextually relevant.

While NFCs can give rise to ISIs through this standard mechanism, I argue that ISIs can also be derived via an independent route—one that does not rely on implicature calculation. Instead, the inference follows directly from the existential presupposition triggered by the NFC. To illustrate, consider the derivation of the ISI in the negated cleft in (22-b), repeated below as (52).

- (52) It's not [every]_F student who has access to their teacher 24 hours a day.
~~ Some students have access to their teacher 24 hours a day.

Negated clefts and other NFCs are restricted to addressing positive QUDs. For example, the sentence *It wasn't my dog that bit the mailman* can serve as a partial answer to the positive question "Whose dog bit the mailman?", as shown in (53-a). Crucially, however, it cannot function as a complete answer to the negative question "Whose dog didn't bite the mailman?", as illustrated in (53-b), nor as a partial answer to the same negative QUD, as in (53-c). A question like "Whose dog didn't bite the mailman?" could instead be answered by a positive cleft with negation embedded in the cleft clause—e.g., *It was my dog that didn't bite the mailman*—where the negation forms part of the presupposed material.¹⁸

- (53) a. A: Whose dog bit the mailman?
B: Well, it wasn't [my]_F dog that bit the mailman, but I don't know whose dog did.
b. A: Whose dog didn't bite the mailman?
B: #It wasn't [my]_F dog that bit the mailman.
c. A: Whose dog didn't bite the mailman?
B: #Well, it wasn't [my]_F dog that bit the mailman, but perhaps other people's dogs didn't bite him either.

¹⁸ Note that a negated cleft can also be used to answer polar questions—whether positive or negative—as illustrated in (i). However, these polar questions are best analyzed as subquestions of the constituent question "Whose dog bit the mailman?", meaning that a complete answer to the superquestion entails the answer to the subquestion (see Groenendijk and Stokhof, 1984; Roberts, 1996). Importantly, by raising such subquestions, the speaker also signals that the superquestion is relevant. By contrast, a simple polar question like "Did your dog bite the mailman?" does not necessarily evoke a superquestion such as "Whose dog bit the mailman?" or "Whom did your dog bite?", since it is compatible with broad focus.

- (i) a. A: Was it [your]_F dog that bit the mailman?
B: No, it wasn't [my]_F dog that bit the mailman.
b. A: Wasn't it [your]_F dog that bit the mailman?
B: No, it wasn't [my]_F dog that bit the mailman.

The restriction on the polarity of the QUD entails that negation cannot be part of the focus alternatives. Consequently, the relevant set of alternatives of (52) is the one given in (54).

- (54) Focus alternatives:
 {Some students have access to their teacher 24 hours a day,
 Most students have access to their teacher 24 hours a day,
 Every student has access to their teacher 24 hours a day.}

In contrast, the alternative set in (55) cannot be derived from (52), as it requires negation to fall within the scope of the focus operator that generates the alternatives—a scope configuration that is unavailable in NFCs.

- (55) Focus alternatives (impossible):
 {It is not the case that some students have access to their teacher 24 hours a day,
 It is not the case that most students have access to their teacher 24 hours a day,
 It is not the case that every student has access to their teacher 24 hours a day.}

The alternative set in (54) corresponds to the positive QUD in (56).

- (56) QUD: Is it some, most, or all students who have access to their teacher 24 hours a day?

The assertion in (57) constitutes a partial answer to the QUD: it rejects the alternative with 'every', but remains compatible with the remaining alternatives. As such, it does not fully resolve the QUD.

- (57) Assertion: It is not the case that every student has access to their teacher 24 hours a day.

As argued in Subsection 4.1.2, a cleft presupposes that the QUD contains a true answer.¹⁹ Accordingly, the negated cleft in (52) presupposes that the disjunction of the alternatives in (54) is true. This yields the existential presupposition in (58).

- (58) Presupposition: Some students have access to their teacher 24 hours a day.

When this presupposition is combined with the assertion in (57), the resulting interpretation is that not every student has access to their teacher 24 hours a day, but some do—exactly the 'not all (but some)' ISI observed in (52). In other words, the inference in (52) is simultaneously an existential inference and an ISI.

The derivation of ISIs in negated QP constructions such as (59) proceeds in essentially the same way, since these constructions also qualify as NFCs, as will be shown in Subsection 5.2.

- (59) Not [every]_F student has access to their teacher 24 hours a day.
 $\rightsquigarrow$ Some students have access to their teacher 24 hours a day.

Note that the proposed analysis of ISIs arising from existential presuppositions relies on the assumption that 'none' ('not some') is not a scalar alternative to 'all'—otherwise, the existential presupposition in (58) would not arise. While there are competing accounts of how scalar alternatives are generated, there is broad consensus on this particular point. According to the lexical approach, scalar alternatives are constrained by monotonicity: positive and

¹⁹ The exhaustivity component of the cleft operator in (49) produces a vacuous presupposition, since the prejacent in (52) is the strongest alternative in the QUD.

negative quantifiers belong to separate scales and cannot serve as alternatives to one another (Horn, 1989; Matsumoto, 1995). As a result, ‘none’ does not appear on the scale of {some, all}. On the structural approach (Katzir, 2007), ‘none’ is likewise excluded as an alternative to ‘all’ because it is structurally more complex—being composed of a negation operator and an existential quantifier. Under this view, alternatives must be structurally simpler or equally complex, which ‘none’ fails to satisfy in relation to ‘all’.

Another crucial assumption is that in (52), focus falls specifically on the quantifier (*every*) rather than on the entire cleft phrase (*every student*). If the entire cleft phrase were focused, the set of alternatives would also include propositions like *Most TAs have access to their teacher 24 hours a day*, where *every student* is replaced by other generalized quantifiers with different restrictors. As a result, the disjunction of these alternatives would no longer yield the existential presupposition in (58).

Far from being a limitation, this is in fact a desirable prediction of the current analysis. When focus does not fall on the quantifier, the resulting focus alternatives are not scalar alternatives of one another, and the existential presupposition that arises is of a different kind—one that does not give rise to an obligatory ISI. This prediction is supported by data from Swedish, a language in which negated clefts are more naturally used to express ‘Not all X are Y’ propositions than in English, where such clefts are relatively restricted in distribution (Amiraz, 2025b). Consider the example in (60). In the given context, focus most naturally falls on the quantifier, yielding a set of alternatives parallel to those in (54). The disjunction of the focus alternatives gives rise to the existential presupposition ‘Some of my savings were invested in that stock’. This presupposition, in turn, leads to an obligatory ISI. Since the ISI contradicts the established common ground, the sentence is infelicitous, mirroring the behavior of the English translation.

(60) Swedish (Alice Bondarenko, p.c.)

Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock’s value plummets to zero.

Bob says:

#Tack och lov att det inte var [alla]_F mina besparingar som var
 thank and praise that it NEG be.PST all.PL my.PL savings which be.PST
 investerade in den där aktien!
 invest.PTCP.PL in that there stock
 ‘Thank goodness not all of my savings were invested in that stock!’

In contrast, the same sentence becomes felicitous in a context that licenses focus on a different part of the cleft phrase. This is shown in (61), where the possessive pronoun bears contrastive focus. In this case, the resulting existential presupposition is that there is someone whose entire savings were invested in the stock—namely, John. Since the existential presupposition does not correspond to the ISI, the scalar inference must instead be derived via implicature calculation. As a result, it is defeasible and may be suspended in this context.

(61) Swedish (Alice Bondarenko, p.c.)

Context: John invested all of his savings in a stock he claimed would be the next Apple. He then tried to persuade his friend Bob to do the same. But before Bob could act on the advice, the stock’s value plummeted to zero.

Bob says:

Tack och lov att det inte var alla [mina]_F besparingar som var
 thank and praise that it NEG be.PST all.PL my.PL savings which be.PST

investerade in den där aktien!
 invest.PTCP.PL in that there stock
 'Thank goodness it's not all of MY savings that were invested in that stock!'

In sum, ISIs triggered by NFCs follow directly from an existential presupposition, but only when the quantifier itself is focused. In other words, NFCs trigger an existential presupposition, which may or may not be identical to the ISI, depending on focus placement. If it is, the ISI becomes non-cancelable.

4.2.2. Projection and local accommodation

Similarly to other presuppositions, presuppositional ISIs triggered by NFCs can either project or be locally accommodated, depending on the context. Consider, for instance, the contrast in (62), where 'not all' appears in the complement of *hope*, an entailment-canceling context. In (62-a), the non-NFC does not trigger an ISI in this context, and the sentence is fully felicitous. By contrast, the NFC in (62-b) gives rise to an existential inference—that some members of the audience did throw tomatoes—and the speaker merely hopes that the rest did not join in. Crucially, this inference does not seem to affect the asserted (bouletic) content of the sentence, which concerns what the speaker wished for. Instead, the existential inference projects from the scope of *hope*.

(62) *Context: After returning home from a disastrous theatrical performance, an actor tells his wife, "Tonight was the worst performance of my life. I was terrified of the audience when I stepped on stage to take the final bows."*

His wife replies:

- a. I hope they didn't all start throwing tomatoes at you!
- b. I hope not all of them started throwing tomatoes at you!

While presuppositions typically project from entailment-canceling contexts, they can also be locally accommodated in certain cases. This was previously illustrated by (31-b), where the ISI is locally accommodated within the antecedent of the counterfactual conditional. This pattern is consistent with how other types of presuppositions affect the interpretation of counterfactuals (Rooth, 1999). Consider the contrast between (63-a) and (63-b). The sentence in (63-a) is natural: the closest possible worlds where Oswald didn't kill Kennedy are presumably those in which Kennedy was not assassinated at all and may well have been re-elected. In (63-b), by contrast, the cleft construction triggers an existential presupposition—that someone killed Kennedy. If this presupposition merely projected globally, the sentence should be as acceptable as (63-a), since the presupposition is satisfied. However, in this context, the presupposition is instead locally accommodated within the antecedent. As a result, the relevant counterfactual situations are restricted to those in which Oswald did not kill Kennedy, but someone else did. The sentence then asserts that in those worlds, Kennedy—despite having been assassinated—is re-elected, rendering the statement incoherent.

- (63) a. If Oswald hadn't killed Kennedy, Kennedy would have been re-elected.
- b. #If it hadn't been Oswald who killed Kennedy, Kennedy would have been re-elected.

The same mechanism accounts for the interpretation of (31-b): the existential presupposition introduced by the NFC is locally accommodated within the antecedent of the counterfactual conditional, rather than projecting globally: it restricts the relevant counterfactual situations to those where some, though not all, of the team members perished in the crash.

5. Remaining issues

5.1. Suspending presuppositional ISIs

A potential counterexample to the projection of ISIs in NFCs is given in (64). As noted by Spector and Sudo (2017), when a weak scalar item appears in the complement of a Strawson-upward-entailing factive predicate, such as *be aware*, it tends not to trigger a scalar inference in its presupposition. For example, (64-a) does not imply that not all of the students are vegan.²⁰ Similarly, the floating quantifier construction in (64-b) does not give rise to an ISI in the presupposition of *aware*. Surprisingly, the negated QP construction in (64-c) behaves the same way and does not trigger a presuppositional ISI either.

- (64) A: Why is John preaching to these students about the health hazards of eating red meat? Doesn't he know that they're all vegan?
 B: I think he's only aware that...
 a. {SOME of them are vegan / SOME of them don't eat meat}.
 b. they don't ALL eat meat—he doesn't seem to realize that none of them do.
 c. not ALL of them eat meat—he doesn't seem to realize that none of them do.

It remains an open question why the ISI can be canceled in (64-c), in contrast to other ISIs triggered by NFCs discussed in this paper. However, a similar pattern is observed in (65), which is relatively acceptable despite the fact that the existential presupposition triggered by the cleft contradicts the common ground.

- (65) John doesn't know that the mailman lied about being bitten. He only knows that it wasn't [his]_F dog that bit him.

The apparent suspension of the existential presuppositions in (64-c) and (65) may be related to a more general pattern of presupposition projection under attitude verbs. When the complement of a non-factive attitude verb presupposes *p*, the sentence as a whole presupposes not *p* itself, but rather that the attitude holder believes *p* (Karttunen, 1974; Heim, 1992). Emotive factive verbs such as *be glad* and *regret*, on the other hand, are usually taken to presuppose that their complement is true, rather than merely believed to be true by the attitude holder. Nevertheless, this presupposition can sometimes be suspended when the context explicitly establishes that the attitude holder is mistaken (Gazdar, 1979; Heim, 1992; Egré, 2008). For example, the sentence *Mary, who was under the illusion that it was Sunday, was glad that she could stay in bed* does not presuppose that Mary could stay in bed, but rather that she believed she could.

In this light, the existential presuppositions triggered by the embedded NFCs in (64-c) and (65) may likewise be attributed to the attitude holder's beliefs. Thus, John is taken to believe that some of the students eat meat, or that someone's dog bit the mailman, even though the speaker does not believe that. This interpretation, however, seems to require contextual support, such as an explicit indication that the attitude holder is mistaken, and perhaps also the presence of *only* in (64-c) and (65).²¹

²⁰ Example (64-a) triggers a DSI in its assertion: it implies that John is not aware that all of the students are vegan. See Spector and Sudo (2017) for further details.

²¹ The predicates in these examples are cognitive factives, namely *be aware* and *know*, rather than emotive factives such as *be glad* and *regret*, and are generally expected to be veridical (Egré, 2008). Indeed, the speaker remains committed to the truth of the complement, but not to the existential presupposition triggered within it. For example, (65) entails that John's dog did not bite the mailman, but it does not entail that some other dog did.

Beyond projection, another standard diagnostic for presuppositions is their resistance to cancelation or suspension when the presupposition trigger appears in an unembedded clause (see Subsection 4.1.2). However, unlike in projection, we do not observe a clear distinction between presuppositional ISIs and ordinary scalar inferences with respect to this diagnostic. For example, the DSI typically associated with ‘some’ or ‘most’ can often be suspended in contexts where the speaker’s ignorance about the stronger alternative (‘all’) is made explicit, as shown in (66-a). Unexpectedly for the present analysis, (66-b)—which involves an NFC—is also acceptable, even though the presuppositional ISI it triggers cannot be accommodated. This suggests that the inference is suspended, contrary to what the presuppositional account would predict.

- (66) a. We don’t know yet whether all of the students passed, but we can already say that {some / most} of them did.
 b. We don’t know yet whether any of the students failed, but we can already say that not all of them did.

In the case of (66-a), suspension of the DSI is expected: under the grammatical approach, the stronger alternative with ‘all’ can be ignored if it is not contextually relevant, while under the pragmatic approach, the implicature does not arise if the speaker’s competence is in doubt. However, neither mechanism should apply in (66-b). Under the presuppositional analysis, this sentence is expected to be just as bad as (67), where an existential presupposition is explicitly claimed not to be part of the common ground.

- (67) #We don’t know yet whether anyone’s dog bit the mailman, but we can already say that it wasn’t [Mary’s]_F dog that bit him.

Thus, it remains an open question why presuppositional ISIs seem to be more easily suspended than other existential presuppositions triggered by the same constructions.

5.2. Negated quantifier phrase constructions are NFCs

In Subsection 2.2, I classified ‘Not all X are Y’ constructions as either NFCs or non-NFCs, depending on whether analogous constructions with a focused DP gave rise to existential presuppositions or implicatures. However, this diagnostic cannot be directly applied to English negated QP constructions such as *Not all of the cookies burned*, since English lacks a parallel constituent negation construction with a focused DP (e.g., **Not the cookies burned*). Other languages, however, do allow constituent negation of a focused DP. I will leverage this fact to argue that negated QP constructions in general should be classified as NFCs. This assumption played a pivotal role in the derivation of ISIs emerging from these constructions, as discussed in Subsection 4.2.

In Hebrew, constituent negation of a focused DP qualifies as an NFC and, like a negated cleft, triggers an existential presupposition (see Subsection 2.1.1). This is illustrated in (9-b), repeated below as (68).

- (68) Hebrew
 lo ha-kelev [šel-i]_F našax et ha-davar.
 NEG the-dog POSS-1SG bit ACC the-mailman
 ‘It wasn’t my dog that bit the mailman.’
 ~> Someone’s dog bit the mailman.

Constituent negation of a focused DP bears a superficial resemblance to a negated QP construction, as seen in (69). The key question is whether (69) shares the same semantics

as (68)—specifically, whether it should also be analyzed as an NFC that triggers an existential presupposition.

- (69) Hebrew
 lo [kol]_F ha-ugiy-ot nisrefu.
 NEG all the-cookie-PL burned
 'Not all of the cookies burned.'

To demonstrate that negated QP constructions in Hebrew are a subtype of constituent negation, I will examine non-scalar inferences that arise when focus is placed not on the quantifier itself, but on its nominal complement. These inferences provide independent evidence that such constructions behave like NFCs.

As a point of comparison, I begin with examples from English involving non-NFCs, where only the nominal complement of the quantifier is focused. In (70), corrective focus highlights alternatives of the form 'I burned all the *NP*'. Because the quantifier *all* is present in every alternative, it remains part of the given material, and no scalar alternatives to 'all' are evoked. As a result, no ISI is triggered.

- (70) *Context: John was baking cookies and muffins. He took the cookies out of the oven on time, but he left the muffins for too long and they all burned. When Bill entered the kitchen, he noticed smoke coming out of the oven.*
 Bill: You burned all the cookies!
 John: No, I didn't burn all the [cookies]_F, I burned all the [muffins]_F, the cookies are fine.

That is, in the context provided in (70), the sentence *I didn't burn all the cookies* does not imply that some of the cookies burned—an inference that would typically arise as an ISI. This is evident from the fact that John's response does not register as a contradiction. Instead, the interaction between negation and free focus gives rise to a different kind of existential implicature: namely, that there exists some type of pastry—other than cookies—of which all the individual items burned.

Now consider analogous examples involving negated QP constructions in Hebrew. As in (70), focus can fall on the nominal complement of the quantifier rather than on the quantifier itself in (71). In this configuration, the resulting existential inference is that there exists a true alternative of the form 'All of the *NP* burned'. Like in (70), the ISI typically associated with 'not all' can be suspended in this context, since it is not directly derived from the existential inference.

- (71) Hebrew
 lo kol [ha-ugiy-ot]_F nisrefu, kol [ha-mafin-im]_F nisrefu. ha-ugiy-ot
 NEG all the-cookie-PL burned all the-muffin-PL burned the-cookie-PL
 yacu beseder.
 come.out.PST.3PL fine
 'It's not the cookies that all burned—it's the muffins that all burned. The cookies are fine.'

However, unlike in (70), the existential inference triggered by a negated QP construction is a presupposition, not a cancelable implicature. This is evidenced by the infelicity of (72), where the existential presupposition is not satisfied.²²

(72)

²² It is worth noting that the English translation of (72), which involves a negated cleft, is similarly infelicitous for the same reason: its existential presupposition is not met.

Hebrew

#ha-ugiy-ot kvar mixuc l-a-tanur, az min ha-stam lo kol
 the-cookie-PL already outside DAT-DEF-oven so from DEF-just NEG all
 [ha-ugiy-ot]_F nisrefu.

the-cookie-PL burned

'The cookies are already out of the oven, so it's obviously not the cookies that all burned. [It's probably just some burnt fat at the bottom of the oven causing the smoke.]'

These examples support the conclusion that negated QP constructions in Hebrew qualify as NFCs. Like negated clefts, they consistently trigger existential presuppositions. When focus falls on the quantifier, the existential presupposition is identical to the ISI, resulting in an obligatory ISI. By contrast, when focus targets the quantifier's nominal complement, the resulting existential presupposition is distinct from the scalar inference. In such cases, while the existential presupposition cannot be canceled, as shown in (72), the ISI can be suspended, as demonstrated in (71), since it arises as an ordinary scalar implicature.²³

Still, one might question whether this line of reasoning extends to languages like English, which lack constituent negation of a focused DP. While this possibility cannot be definitively ruled out, I argue that analyzing negated QP constructions in English as NFCs can offer insight into an otherwise puzzling constraint on focus placement. Specifically, I observe that unlike in Hebrew, negated QP constructions in English require focus to fall on the quantifier itself. This is evidenced by the infelicity of (73), contrasting with the acceptability of the parallel example in Hebrew (71), as well as the English non-NFC in (70).

(73) Bill: All the cookies burned!

John: #No, not all the [cookies]_F burned—all the [muffins]_F burned. The cookies are fine. (Todd Snider, p.c.)

Example (73) cannot be interpreted as involving corrective focus on *cookies*, since the construction does not permit focus to be restricted to the nominal complement. Due to this constraint on focus placement, negated QP constructions in English necessarily evoke scalar alternatives, which results in obligatory ISIs.

Why, then, does English differ from Hebrew in this respect? I suggest that the explanation likely lies in the availability of constituent negation in Hebrew. As shown in (69), Hebrew permits constituent negation of a focused DP, whereas English does not (e.g., **Not the cookies burned*). This syntactic difference constrains the focus options available in English negated QP constructions. At a descriptive level, the inability of constituent negation in English to associate with a focused DP inside a QP appears to reflect a broader constraint: constituent negation cannot associate with a focused DP at all in a free-standing clause, even when that DP is linearly adjacent to the negative particle. In other words, constituent negation in English requires a focused quantifier for licensing. By contrast, Hebrew constituent negation is more flexible. It can associate not only with an adjacent focused DP, but also with a non-adjacent focused element inside the negated DP, as shown in (68). This mirrors the focus flexibility of negated clefts, as illustrated in (48-b) and (48-c). It is therefore unsurprising that Hebrew also permits constituent negation to associate with a focused DP inside a QP, as seen in (71).

This cross-linguistic variation may be attributed to syntactic differences, such that constituent negation is subject to different selectional restrictions across languages. However,

²³ Notably, the interpretive flexibility of negated QP constructions in Hebrew mirrors that of negated clefts in Swedish (60)–(61), discussed earlier in Subsection 4.2.1.

a semantic explanation is also potentially available. Specifically, it could be the case that English constituent negation is associated with a stronger presupposition compared to its Hebrew counterpart.

In Hebrew, constituent negation, like clefts, presupposes that the alternative set contains a true alternative (see Subsection 4.1.2). This condition can be satisfied whether focus falls on the quantifier or on its nominal complement. As a result, although the default reading places focus on the quantifier, the quantifier can be deaccented in the right discourse context, as illustrated in (71).

In English, on the other hand, constituent negation seems to impose a stricter constraint. According to Wu (2025), the construction not only presupposes the existence of a true alternative but must also entail a specific alternative in the alternative set. This condition can be satisfied in one of two ways. The first way is when one of the alternatives is entailed by all the others, in which case the disjunction of the alternative set corresponds to the weakest member of the set. This is the case when focus falls on the quantifier, as in (74), where the alternative with ‘some’ is the weakest. The sentence is acceptable as a free-standing clause because there is no need to explicitly assert a specific alternative—the alternative with ‘some’ is already entailed.

(74) Not [all]_F the cookies burned.

Focus alternatives: {Some of the cookies burned, Most of the cookies burned, All of the cookies burned}

The second way is when the sentence is paired with an explicit contrasting alternative. If focus falls on the nominal complement, as in (75), the disjunction of the alternatives does not entail any specific member. As a result, the sentence is infelicitous as a stand-alone clause. However, the construction can still be licensed if a contrasting alternative is explicitly asserted in a corrective *but*-clause, as in (76).

(75) #Not all the [cookies]_F burned.

Focus alternatives: {All of the cookies burned, All of the muffins burned, All of the croissants burned, ...}

(76) Not all the [cookies]_F, but all the [muffins]_F, burned.

In sum, the comparison between English and Hebrew constituent negation suggests that the restrictions on focus placement in English are best explained if English negated QP constructions are analyzed as NFCs—constructions that must associate with narrow focus.²⁴

5.3. Other obligatory scalar inferences

The central aim of this paper has been to explain why NFCs generate obligatory ISIs. I have assumed throughout the paper that such obligatory inferences are exceptional, since scalar inferences are normally cancelable. Yet there are contexts in which “ordinary” scalar inferences resist cancelation, which might challenge the assumption that cancelable ISIs triggered by non-NFCs represent the norm, whereas those triggered by NFCs are exceptional.

In this subsection, I discuss two cases where scalar inferences tend to be obligatory:

(i) when the assertion is contextually equivalent to a logically stronger alternative (Magri, 2009, 2011), known as a Magri effect, and (ii) when a weak scalar item is embedded in the complement of a factive SDE predicate (Spector and Sudo, 2017).

²⁴ For an alternative explanation of the narrow-focus constraint on negated QP constructions, see Zeijlstra (2022a, 168–170).

5.3.1. Magri effects

Magri (2009, 2011) observes that sentences involving weak scalar items, such as ‘some’, are infelicitous in contexts where the assertion is contextually equivalent to a logically stronger scalar alternative. For example, in (77), the weaker statement in (77-a) is equivalent to the stronger alternative in (77-b), given the contextual assumption that Prof. Smith always assigns the same grade to all of his students. This poses a challenge for pragmatic approaches to scalar inferences: since the alternative with ‘some’ is just as informative as the alternative with ‘all’ in this context, no pragmatic inference is expected to arise.

- (77) *Context: Prof. Smith always assigns the same grade to all of the students in his class.* (Magri, 2011, 2)
- a. #This year, Prof. Smith gave an A to some of his students.
 $\rightsquigarrow$ Prof. Smith gave an A to some, but not all, of his students.
 - b. Relevant alternative: This year, Prof. Smith gave an A to all of his students.

According to Magri’s account, a covert EXH operator is obligatorily present in both matrix and embedded clauses. On this view, scalar inferences are only optional when stronger alternatives are deemed irrelevant, allowing them to be pruned from the domain of EXH. However, when an alternative is contextually equivalent to the prejacent of EXH, it is necessarily considered relevant and must be excluded. In (77-a), this yields the inference that Prof. Smith did not assign an A to all of his students, contradicting the common ground.

Magri effects tend to be less robust in ISIs (Amiraz, 2022; Sudo, 2025). This is illustrated by (78): the logically weaker alternative in (78-a) is contextually equivalent to the scalar alternative in (78-b). Still, both alternatives are felicitous.

- (78) *Context: Prof. Smith always assigns the same grade to all of the students in his class. This year, all of the students did very poorly, except for Jane, whose work truly deserved an A. Prof. Smith reasons that it is better that ten undeserving students pass than that one deserving student fail, and so he decides to give everyone a C.* Prof. Smith (speaking to a colleague):
- a. It is only because of Jane that I didn’t give all the students an F.
 $\rightsquigarrow$ It is only because of Jane that I didn’t give all, but only some, of the students an F.
 - b. Relevant alternative: It is only because of Jane that I didn’t give any of the students an F.

Similar examples can be constructed with matrix clauses. For instance, in the context of (78), the students could felicitously say:

- (79) Thanks to Jane, we didn’t all get an F!

Another type of context in which Magri effects sometimes arise is one where the common ground already entails the stronger alternative, making that alternative trivially relevant (Marty and Romoli, 2021, 254), according to the notion of relevance in Lewis (1988). Here, too, there seems to be a contrast between DSIs and ISIs triggered by non-NFCs. For instance, the felicity of (5-a), repeated here as (80-a), contrasts with the infelicity of (80-b), which gives rise to a DSI in the same context.

- (80) *Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock’s value*

plummets to zero.

Bob says:

- a. Thank goodness I didn't invest all of my savings in that stock!
- b. #Thank goodness I kept {some / most} of my savings in my regular savings account!

It should be noted, however, that the absence of Magri effects with certain ISIs is not necessarily incompatible with Magri's theory of obligatory exhaustification. In fact, Magri (2011, 38–40) discusses a contrast similar to the one between (81-a) and (81-b). While (81-a) is felicitous and does not trigger an ISI, (81-b) appears to give rise to an obligatory ISI.

- (81) *Context: Prof. Smith always assigns the same grade to all of the students in his class. This year, he gave them all a C.*
- a. It is not the case that all of the students got an A.
 - b. #Not all of the students got an A.

Magri proposes that the acceptability of (81-a) can be explained by positing two instances of the EXH operator, as in (82). In this structure, the embedded EXH has no local strengthening effect, since the scalar alternative with 'some' (i.e., 'some of the students got an A') is weaker than the prejacent of EXH (i.e., 'all of the students got an A') and therefore cannot be excluded. However, the embedded EXH plays a crucial role in the computation of alternatives to the prejacent of the matrix EXH. Specifically, the relevant alternative involves replacing 'all' with 'some', yielding the proposition: 'It is not the case that some, but not all, of the students got an A', which means that either none or all of the students got an A. Importantly, this alternative is not contextually equivalent to the prejacent of the matrix EXH (i.e., 'It is not the case that all of the students got an A'), since only the former is true in a scenario where all the students got an A. Because of this lack of equivalence, the alternative may be pruned from the domain of EXH if it is deemed irrelevant, and an ISI is not obligatory.

- (82) EXH [It is not the case that [EXH [all of the students got an A]]].

In contrast, in (83), there is no embedded EXH operator inserted between the negation and the QP, because there is no syntactically appropriate position for such an operator.²⁵ Since no embedded EXH is present, the scalar alternative with 'none' is contextually equivalent to the prejacent of the matrix EXH operator. As a result, this alternative must be excluded, giving rise to a contextual contradiction.

- (83) EXH [Not all of the students got an A].

Importantly, Magri's analysis may also explain the contrast argued for in this paper between negated QP constructions and non-NFCs. This follows if only the latter allow an embedded EXH operator to be inserted between negation and the quantifier. Although this approach is unlikely to extend to the other types of NFCs discussed in this paper (see Section 3), and therefore does not achieve the same empirical coverage as the analysis proposed here, it is nevertheless important to assess its viability even in the more limited domain of the contrast between negated QP constructions and clausal negation in English.

²⁵ Magri (2011, 39) analyzes *not all* as a complex determiner, yielding the structure *[[not all] the students] got an A*. By contrast, the present paper adopts the view that negation modifies the entire QP, resulting in the structure *[not [all the students]] got an A* (see Collins 2020 for arguments in support of this structure). Nevertheless, even under a negated QP analysis, it is plausible to assume that EXH cannot syntactically intervene between negation and the QP.

I argue that Magri's analysis faces several significant challenges. First, Fox and Spector (2018, 35) argue that a higher EXH operator cannot associate with the focus value of a scalar item that has already been targeted by a lower EXH operator, due to a general constraint on association with focus (Rooth, 1985, 1992). Instead, they propose that the relevant alternative for the higher EXH operator is derived by deleting the lower EXH. Crucially, for Magri's analysis of (82) to succeed, one must assume not only that the scalar item's focus value remains available to both instances of EXH, but also that the alternative obtained by deleting the lower EXH is unavailable. Otherwise, this alternative would be contextually equivalent to the prejacent of the higher EXH and therefore could not be pruned. Magri's account thus appears to rely on non-standard assumptions regarding alternatives.

Second, as pointed out by Chatain (2025), assuming an embedded EXH below negation appears to make incorrect predictions for weak scalar items. Consider the scale ⟨allowed, required⟩. When unembedded, the sentence with the weak scalar item is infelicitous due to an obligatory DSI (a Magri effect), as shown in (84).

- (84) *Context: In a fictional country, citizens are legally obliged to wear every item listed in the official dress code, and no others.*
- a. #Citizens are allowed to wear hats.
 - b. Citizens are required to wear hats.

Under negation, however, the pattern reverses: the sentence with the weak scalar item is felicitous, whereas the one with the strong scalar item is infelicitous (though perhaps less so than (84-a)), as shown in (85). If one assumes an embedded EXH below negation, as in (86), the opposite pattern is predicted. Since *allowed to wear bracelets* and *required to wear bracelets* are contextually equivalent, the embedded EXH obligatorily derives the scalar inference. Consequently, the prejacent of the higher EXH becomes 'Citizens are either forbidden from or required to wear bracelets'. The alternative with *required* is no longer contextually equivalent to this prejacent. If it is excluded, the resulting meaning is 'Citizens are required to wear bracelets'; if it is instead pruned, the meaning remains disjunctive. Neither outcome matches the observed interpretation.

- (85) a. Citizens are not allowed to wear bracelets.
b. #Citizens are not required to wear bracelets.
- (86) EXH [Citizens are not [EXH [allowed to wear bracelets]]]

Finally, I observe that Magri effects do not always arise with DSIs either. For instance, in (87), the weaker alternative in (87-a) is not only felicitous, but even preferred over the stronger alternative in (87-b). The reason is that the interjection *phew!* expresses relief about the present situation, which accords with the speaker's desires. In this context, the relevant desire-worlds are presumably worlds in which citizens are free to choose what to wear. The stronger alternative in (87-b), by contrast, suggests that the speaker is relieved that the citizens are required to wear shoes, rather than that they are not longer forbidden from wearing them. This appears to explain why (87-a) can be used without giving rise to a DSI.

- (87) *Context: In a fictional country, the official dress code is updated every day. Citizens are legally obliged to wear every item listed in the dress code for that day, and no others. Yesterday, the dress code required everyone to go barefoot. Today, it requires everyone to wear shoes.*
- A man reads today's dress code and says:

- a. Phew! We're allowed to wear shoes today.
 ✗ We're not required to wear shoes today.
- b. Relevant alternative: Phew! We're required to wear shoes today.

In sum, it appears that not only ISIs, but also certain DSIs, can sometimes be suspended in contexts where they contradict the common ground. However, there seems to be a distinction within DSIs between weak scalar items such as 'allow' in (87-a) and intermediate scalar items such as 'most' in (80-b). Curiously, 'some' seems to pattern with intermediate scalar items in this respect, as illustrated by (80-b) and as will be further shown in the next subsection. Although the source of this distinction remains an open question, my working assumption is that DSIs are generally not obligatory, much like ISIs triggered by non-NFCs.

5.3.2. Scalar inferences in factive SDE environments

Spector and Sudo (2017) argue that DSIs are obligatory in factive SDE environments, such as the complements of predicates like *unaware* or *surprised*. At first glance, this poses a challenge for my account: in these environments, ISIs triggered by non-NFCs remain suspendable, as seen in (3-a), whereas the obligatory ISIs of NFCs persist (3-b). This makes ISIs triggered by NFCs appear to pattern with DSIs, casting doubt on the claim that they constitute a distinct phenomenon.

To address this issue, some background on DSIs in factive SDE environments is in order. Sentences featuring a weak scalar item within a factive SDE environment often give rise to a special kind of embedded DSI, in which the scalar term receives an upper-bounded interpretation in the presupposition, but not in the assertion (Russell, 2006; Simons, 2006; Gajewski and Sharvit, 2012). For instance, the sentence in (88) strongly implies that the speaker (not John) believes that not all of the students smoke—an inference that reflects an embedded DSI. Crucially, however, this inference applies only to the presupposed content. If 'some' were interpreted as 'some but not all' in the asserted meaning, the sentence would mean that John believes that either none or all of the students smoke (the result of negating 'some but not all'). Yet, (88) is naturally understood to mean that John believes that none of the students smoke—i.e., the negation of the plain (i.e., not enriched) meaning of 'some'. This suggests that the DSI is confined to the presupposition and does not contribute to the asserted content.

- (88) John is unaware that some of the students smoke. (Spector and Sudo, 2017)
- a. Assertion: John doesn't know whether some of the students smoke.
 - b. Presupposition: Some, but not all, of the students smoke.

Sentences like (88) exhibit an additional notable property: they tend to be infelicitous when a scalar alternative has a stronger presupposition, and that presupposition is entailed by the common ground (Spector and Sudo, 2017). For example, (89-a), in which the weak item 'some' appears in the scope of the factive SDE predicate *unaware*, is infelicitous in the context of (89). The infelicity of (89-a) stems from competition with the alternative with 'all' (89-b), which carries a stronger presupposition entailed by the common ground. Put differently, (89-a) triggers an obligatory DSI in its presupposition: it implies that not all of the students smoke, contradicting the established common ground.²⁶

²⁶ Spector and Sudo (2017) analyze this as a "presupposed ignorance inference" rather than a scalar inference. That is, the presupposition of (89-a) is that the speaker does not know whether all of the students smoke. However, following Marty and Romoli (2021) and Bassi et al. (2021), I treat this inference as a scalar inference for present purposes.

- (89) *Context: John wonders if any of the students smoke. We know that all of them do.* (Spector and Sudo, 2017, 477)
- a. #John is unaware that some of the students smoke.
 - b. Relevant alternative: John is unaware that all of the students smoke.

A natural question arises at this point: are the infelicity effects observed with 'some' in factive SDE environments the same as those observed with 'not all' in NFCs? For example, both (90-a) and (90-b) are infelicitous in the context of (3), repeated here as (90).

- (90) *Context: Mary teaches Intro to Statistics. She accidentally gave the students the exam for Advanced Statistics. She expected that they would all fail, but surprisingly, they all passed.*
- Mary (to her students): You should all be proud of yourselves for passing the exam. Frankly, ...
- a. #Frankly, I was surprised that some of you passed!
Presupposition: Some (but not all) of you passed.
 - b. #Frankly, I was surprised that not all of you failed!
Presupposition: Not all (but some) of you failed.

By contrast, suspending an ISI triggered by a non-NFC is possible in this context. For example, (91-a) is felicitous, even though (91-b) has a stronger presupposition, which is entailed by the common ground.²⁷

- (91) *Context: same as (90)*
- a. I was surprised that you didn't all fail!
Presupposition: It is not the case that all of you failed (possibly none did).
 - b. Relevant alternative: I was surprised that none of you failed!

Despite this similarity between DSIs and ISIs triggered by NFCs, I maintain that the underlying phenomena are fundamentally distinct. I argue that ISIs triggered by non-NFCs represent the norm, whereas those triggered by NFCs constitute the exception. First, DSIs in factive SDE environments can often be canceled by contrastive focus, whereas ISIs triggered by NFCs cannot be suspended in this manner. For example, in (92), the alternative with the weaker presupposition becomes felicitous when the focus particle *even* is added (Spector and Sudo, 2017, 500).

- (92) *Context: same as (89) (Spector and Sudo, 2017, 500)*
- John is even unaware that *SOME* of the students smoke.

I observe that with most weak scalar items, the DSI can also be suspended without the presence of *even*, typically by means of contrastive focus on the scalar item. For example, disjunctions of the form 'either A or B' do not necessarily give rise to the inference 'not both A and B' in these environments. In (93-a), it is not presupposed that only one of 333 Magazine or Nathan Barley existed. Likewise, (93-b) (from the Corpus of News on the Web (NOW), (Davies, 2016)) is felicitous despite the fact that both sensitive and classified documents were found in a search of Mike Pence's home.²⁸

²⁷ Example (91-b) is a scalar alternative to (91-a) because *none* is a lexical realization of 'not some'.

²⁸ In a different news report, Pence's attorney is quoted using a slightly different construction, in which *unaware* takes a prepositional rather than a clausal complement.

- (i) "Vice President Pence was unaware of the existence of sensitive or classified documents at his personal residence," Jacob wrote. [Gangel, Jamie, and Herb, Jeremy, and

Shoreditch is overrun with people apparently blissfully unaware that either 333 Magazine, or Nathan Barley ever existed. [web example²⁹]

- b. The Vice President was unaware that either potentially sensitive or classified documents were transported to his home. [NOW: US; 2017]

These examples suggest that DSIs can indeed be suspended under certain conditions, much like ISIs triggered by non-NFCs (91-a). Nevertheless, there is a discernible difference in how readily the scalar inference can be suspended. DSIs typically require contrastive focus or an overt *even* to be canceled, whereas ISIs triggered by non-NFCs may be naturally suspended without either of these. This is illustrated in (94): (94-a) sounds at least as natural as (94-b), even though the latter carries a stronger presupposition already entailed by the common ground. The naturalness of (94-a) suggests that its ISI is relatively weak and easily suspendable. Accounting for this apparent difference between DSIs and ISIs triggered by non-NFCs is left for future research.

(94) *Context: John bought ten Bitcoins in 2011 for \$1 each. He sold all of them in 2017 when they reached \$1,000 each. After the value soared to \$50,000 per Bitcoin, John wished he still had all of his Bitcoins.*

- a. John regrets not keeping all of his Bitcoins.
Presupposition: It is not the case that John kept all of his Bitcoins (possibly he kept none).
- b. Relevant alternative: John regrets not keeping any of his Bitcoins.

A second argument for treating obligatory ISIs as exceptional is that DSIs cease to be obligatory in negated factive SDE environments, while ISIs triggered by NFCs remain non-cancelable. For instance, while examples (90-a) and (90-b) may seem parallel, the symmetry breaks down when the matrix clause is negated. Specifically, (95-a) becomes felicitous, indicating that 'some' no longer triggers a DSI in that context.³⁰ Similarly, the non-NFC in (95-b) allows the ISI to be suspended, making it likewise felicitous. In contrast, (95-c) remains infelicitous due to the persistence of the 'not all (but some)' ISI.

(95) *Context: Mary expected that 40-70% of the students would pass. Surprisingly, they all failed.*

Mary (to her students):

- a. I wasn't surprised that SOME of you failed, but I certainly did not expect that ALL of you would!
- b. I wasn't surprised that you didn't ALL pass, but I certainly did not expect that NONE of you would!
- c. #I wasn't surprised that not ALL of you passed, but I certainly did not expect that NONE of you would!

The felicity of (95-a) is predicted by Spector and Sudo (2017), given that the global environment becomes Strawson-upward-entailing due to the matrix clause negation.

Perez, Evan. CNN, 24 January 2023, <https://edition.cnn.com/2023/01/24/politics/pence-classified-documents-fbi/index.html>. Accessed on 18 April 2025.]

²⁹ Anthony, James. *The Guardian*, 1 December 2006, www.theguardian.com/artanddesign/artblog/2006/dec/01/deckthewallsbanksygetsfses. Accessed on 20 April 2023.

³⁰ Example (95-a) allows for a reading in which *some* refers to a specific subset of students, taking wide scope over negation. However, the sentence remains felicitous even when this reading is unavailable—for instance, if Mary has no familiarity with individual students and her expectations about pass rates are based solely on statistics from previous years.

However, their account does not explain why (95-c) remains infelicitous. According to the analysis proposed in this paper, on the other hand, the ISI in (95-c) arises from an existential presupposition triggered by the NFC, and, as such, is not sensitive to the monotonicity of the embedding environment.

Finally, and most importantly, NFCs trigger obligatory ISIs even in non-factive environments, such as antecedents of indicative (30-b) and counterfactual (31-b) conditionals, and the complement of *hope* (62-b). In these contexts, both ISIs triggered by non-NFCs and most DSIs are optional. Taken together, these arguments support the view that obligatory ISIs are exceptional and should be treated as a distinct phenomenon.

6. Conclusion

This study proposed a distinction between two types of indirect scalar inferences (ISIs), based on their robustness and context dependence. The findings indicate that the properties of an ISI are intrinsically tied to the focus sensitivity of the negative construction in which it originates. Specifically, negative constructions that are not conventionally focus-sensitive (non-NFCs) generate context-dependent ISIs, which are derived by implicature calculation. Conversely, negative focus constructions (NFCs), which are conventionally focus-sensitive, give rise to robust ISIs that remain impervious to contextual considerations. Unlike other scalar inferences, ISIs emerging from NFCs stem from an existential presupposition triggered by the construction itself.

This classification of ISIs opens new avenues for future research, particularly in the area of pragmatic processing. Although prior work has investigated the processing of ISIs (Cremers and Chemla, 2014; Romoli and Schwarz, 2015; Marty et al., 2020), it has not distinguished between ISIs generated by NFCs and those arising from non-NFCs. As a result, an intriguing question that remains to be addressed is whether these two types of ISIs also differ in terms of their processing.

Abbreviations

ACC = accusative, ADE = adessive, ASP = aspect, CLF = classifier, CNG = connegative, CNTR = contrastive marker, COMP = complementizer, COND = conditional, CONN = connective, COP = copula, DAT = dative, DE = Mandarin Chinese *de*, DECL = declarative, DEF = definite, DEM = demonstrative, ELA = elative, ESS = essive, F = feminine, ILL = illative, LOC = locative, NEG = negation, NMLZ = nominalizer, NOM = nominative, PART = partitive, PASS = passive, PL = plural, POSS = possessive, PRS = present, PST = past, PTCP = participle, SBJV = subjunctive, SG = singular

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Appendix: Cross-linguistic data

This appendix provides additional details on the cross-linguistic data discussed in Section 3, which aimed to test the empirical generalization in (7) regarding the relationship between the type of negative construction and the robustness of the ISI it gives rise to.

The data span nine languages in addition to English: Finnish, French, Hebrew, Japanese, Korean, Mandarin Chinese, Bosnian-Croatian-Serbian, Swedish, and Turkish. The examples are not identical across languages, since certain constructions do not translate directly due to grammatical differences.

For each language, the following information is provided: (i) how standard negation is expressed; (ii) how propositions of the form 'Not all X are Y' are typically realized, and whether the relevant construction qualifies as an NFC or non-NFC; and (iii) illustrative examples showing the type of ISI generated by each construction.

Finnish

In Finnish, standard negation is expressed by an inflecting negative auxiliary, which precedes a non-finite lexical verb (Vilkuna, 2015, 458). While the negative auxiliary typically follows the subject, as in (96-a), it may also appear clause-initially in certain contexts, as illustrated in (96). Clause-initial negation is pragmatically marked and tends to signal the rejection of an explicit or implicit assumption (Vilkuna, 2015, 472–473).

(96) Finnish (Vilkuna, 2015, 472–473)

- a. Anna-lla ei ole koir-a.
Anna-ADE NEG-3SG be.CNG dog-PART
'Anna does not have a dog.'
- b. Ei Anna-lla ole koir-a.
NEG.3SG Anna-ADE be.CNG dog-PART
'Anna does not have a dog.'

When the subject is a universal QP, both negation patterns remain available, as shown in (97). Given the marked status of clause-initial negation, the inverse-scope construction in (97-a) is the pragmatically neutral way of expressing 'Not all X are Y' propositions in Finnish (Maria Vilkuna, p.c.).

(97) Finnish (Maria Vilkuna, p.c.)

- a. Kaikki-lla ei ole koir-a.
everyone-ADE NEG.3SG be.CNG dog-PART
'Not everyone has a dog.'
- b. Ei kaikki-lla ole koir-a.
NEG.3SG everyone-ADE be.CNG dog-PART
'Not everyone has a dog.'

Both constructions in (97) are based on standard negation and permit broad focus. Accordingly, they are classified as non-NFCs. In line with the generalization in (7), these non-NFCs do not consistently trigger ISIs. This is confirmed by the examples in (98-a) and (98-b), where the negative auxiliary either precedes the quantifier (and cliticizes to the

complementizer) or follows it in an inverse-scope configuration. Both are felicitous in the given context (Matti Miestamo, p.c.), indicating the absence of an obligatory ISI.

(98) Finnish (Matti Miestamo, p.c.)

Context: Mary teaches Intro to Statistics. She accidentally gave the students the exam for Advanced Statistics. She expected that they would all fail, but surprisingly, they all passed.

Mary (to her students): You should all be proud of yourselves for passing the exam. Frankly, ...

- a. Suora-an sano-ttu-na ol-i-n yllätty-nyt sii-tä,
 frank-ILL say-PTCP.PASS-ESS be-PST-1SG be.surprised-PTCP.PST it-ELA
 ett=e-tte te kaikki epäonnistu-nee-t!
 COMP=NEG-2PL 2PL all fail-PTCP.PST-PL
 'Frankly, I was surprised that you didn't all fail!'
- b. Suora-an sano-ttu-na ol-i-n yllätty-nyt sii-tä,
 frank-ILL say-PTCP.PASS-ESS be-PST-1SG be.surprised-PTCP.PST it-ELA
 että te kaikki e-tte epäonnistu-nee-t!
 COMP 2PL all NEG-2PL fail-PTCP.PST-PL
 'Frankly, I was surprised that you didn't all fail!'

French

In Standard French, standard negation is expressed using a bipartite construction in which the clitic *ne* precedes the finite verb, and the particle *pas* follows it (Miestamo, 2005, 62).

(99) French (Miestamo, 2005, 62)

Jean ne vient pas.
 Jean NEG come.PRS.3SG NEG
 'Jean is not coming.'

Propositions of the form 'Not all X are Y' are typically expressed either through an inverse-scope construction (100-a) or a floating quantifier construction (100-b) (Amiraz, 2025b). Both are based on standard negation and allow for broad focus, and are therefore classified as non-NFCs.

(100) French

- a. Tous mes amis ne sont pas allés au dernier
 all.PL.M my.PL friends NEG be.PRS.3PL NEG go.PTCP.PL to.the last
 film de Marvel.
 film of Marvel
 'Not all my friends went to the last Marvel movie.'
- (Hemforth and Konieczny, 2019, 169)
- b. Mes amis ne sont pas tous allés au dernier
 my.PL friends NEG be.PRS.3PL NEG all.PL.M go.PTCP.PL to.the last
 film de Marvel.
 film of Marvel
 'My friends did not all go to the last Marvel movie.'

Consistent with the generalization in (7), the inverse-scope construction is felicitous in the given context, as shown in (101-a). By contrast, the floating quantifier construction in (101-b) is perceived as infelicitous, likely due to the presence of an ISI. This is surprising,

since the construction is not an NFC. However, its acceptability improves with a brief pause after *pas* and heavy pitch accent on *toutes* (Angèle Bernard and Keny Chatain, p.c.), suggesting that the ISI may be suspended under certain prosodic conditions.

(101) French (Angèle Bernard and Keny Chatain, p.c.)

Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value plummets to zero.

Bob says:

- a. Heureusement que toutes mes économie-s n'-ont
 fortunately COMP all.F.PL 1SG.POSS.PL saving-PL NEG-have.PRS.3PL
 pas été investies dans ces action-s!
 NEG be.PTCP.PST invest.PTCP.PST.F.PL in DEM.PL stock-PL
 'Thank goodness that my savings weren't all invested in these stocks!'
- b. #Heureusement que mes économie-s n'-ont pas
 fortunately COMP 1SG.POSS.PL saving-PL NEG-have.PRS.3PL NEG
 toutes été investies dans ces action-s!
 all.F.PL be.PTCP.PST invest.PTCP.PST.F.PL in DEM.PL stock-PL
 'Thank goodness that my savings weren't all invested in these stocks!'

Furthermore, a floating quantifier construction does not trigger an ISI in those non-factive environments where scalar inferences are typically not generated. This is illustrated in (102), where this construction is embedded within the complement of *espérer* 'hope', and no ISI is generated. This contrasts with an NFC in English, such as *I hope not all of them start throwing tomatoes at us!*, which presupposes that some audience members will throw tomatoes in any case.

(102) French (Keny Chatain, p.c.)

Context: After a lackluster theatrical performance, as the actors prepare for their final bows, one actor, sensing the audience's mood, whispers to his fellow performers:

J'-espère qu'ils ne vont pas tous nous jeter
 1SG-hope.PRS.1SG COMP-3M.PL NEG go.PRS.3PL NEG all.M.PL 1PL throw-INF
 des tomate-s!
 INDF.PL tomato-PL
 'I hope they don't all start throwing tomatoes at us!'

For these reasons, I do not consider (101-b) a genuine counterexample to the proposed generalization.

Hebrew

In Hebrew, standard negation is expressed by the negative particle *lo*, which precedes the finite verb.

(103) Hebrew (Miestamo, 2005, 286)

hu lo mevašel.
 he NEG cook.PRS.SG.M
 'He doesn't cook.'

Propositions of the form 'Not all X are Y' are usually expressed using a negated QP construction, as in (104-a). In certain contexts, however, two additional strategies are

available: (i) a standard negation construction with verb-subject (VS) word order, illustrated in (104-b), which is licensed only under specific syntactic conditions (Shlonsky and Doron, 1992; Borer, 1995); and (ii) an inverse-scope construction involving standard negation, permitted only when the entire clause is embedded in an SDE environment (Amiraz, 2019), as illustrated in (105-a). Both of these constructions are considered non-NFCs, since they are based on standard negation.

(104) Hebrew

- a. lo kol ha-student-im higiu etmol.
NEG all the-student-PL arrive.PST.3PL yesterday
'Not all of the students came yesterday.'
- b. etmol lo higiu kol ha-student-im.
yesterday NEG arrive.PST.3PL all the-student-PL
'Yesterday, not all of the students came.'

Example (36-a), which features standard negation with VS word order, does not trigger an ISI and is felicitous in the given context. In contrast, the negated QP construction in (36-b) gives rise to a mandatory 'not all (but some)' ISI because it is an NFC, leading to infelicity due to contextual contradiction.

Similarly, example (105-a), which involves an inverse-scope construction with standard negation, lacks an ISI. By contrast, the corresponding negated QP construction in (105-b) triggers an ISI, rendering the sentence almost incomprehensible.³¹

(105) Hebrew

- a. im kul-xem lo sotmim tox šaloš šniyot, atem metim.
if all-2PL.M NEG shut.up.PRS.PL.M in three seconds you dead.PRS.PL.M
'If you don't all shut up in three seconds, you're dead.'
- b. #im lo kul-xem sotmim tox šaloš šniyot, atem metim.
if NEG all-2PL.M shut.up.PRS.PL.M in three seconds you dead.PRS.PL.M
'If not all of you shut up in three seconds, you're dead.'

Japanese

In Japanese, standard negation is expressed by the negative suffix *-na* (Miestamo, 2005, 294), as shown in (106).

(106) Japanese (Miestamo, 2005, 294)

tabe-na-katta
eat-NEG-PST
'I/you/[...] did not eat'

Another negative construction is *wake de wa nai*, which roughly translates as 'it is not the case that'. Structurally, this construction consists of an embedded clause that serves as the complement of the nominalizing noun *wake* 'conclusion, reason, cause', while negation targets the main clause copula. According to Yoshimura (2013), *wake de wa nai* is typically used to object to an inference made by the interlocutor (107-a), or to reject an implication potentially arising from a previous statement (107-b). Because of its discourse function, this construction is typically infelicitous in a standalone sentence, outside the context of a broader discourse.

(107)

³¹ Obligatory ISIs appear to be especially disruptive in conditional warnings like (105-b), possibly more so than in other types of conditionals. See Subsection 2.3 for similar examples in English.

Japanese (Yoshimura, 2013, 49-50)

- a. *Context: Taro was assigned to an overseas branch of his company and asked his girlfriend, Hanako, to come with him, but she refused. Taro asked Hanako why she wanted to stay in Tokyo, and she replied:*

Tokyo ni nokori-tai wake-de-wa na-i no. Issho ni ike
Tokyo in stay-want conclusion-COP-CNTR NEG-PRS PRT together go.HYP
na-i dake na no.

NEG-PRS only COP PRT

'It is not that I want to stay in Tokyo. I just can't go with you.'

- b. Suzuko wa jimi de tan-tan to shite i-ru keredo, yowaki toka
Suzuko TOP plain and calm do but weak-spirited or
uchiki na wake-de-wa na-i.
bashful conclusion-COP-CNTR NEG-PRS
'Suzuko is plain and calm, but it does not follow that she is weak-spirited or bashful.'

According to Amiraz (2025b), *wake de wa nai* also serves a distinct function: marking wide-scope negation over a preceding scope-bearing element, such as a subject QP, as in (108). Crucially, the sentence in (108) is not subject to the discourse felicity conditions typically associated with *wake de wa nai*. Instead, it is pragmatically neutral and fully acceptable as a standalone utterance. In fact, it is the most natural way of expressing 'Not all X are Y' propositions. Therefore, I assume that, unlike in (107), the construction associates with narrow focus on the QP in (108).

(108) Japanese (Amiraz, 2025b)

Seito-tati-no zenin-ga siken-ni uka-tta wake-de-wa na-i.
student-ASS-GEN all-NOM exam-DAT pass-PST conclusion-COP-CNTR NEG-PRS
'Not all of the students passed the exam.'

(lit. 'It is not the conclusion that all of the students passed the exam.')

Strictly speaking, *wake de wa nai* does not qualify as an NFC, since it is usually compatible with broad focus, as seen in (107). Indeed, in unembedded clauses, *wake de wa nai* seems to behave like a typical non-NFC: it generally gives rise to an ISI, as in (108), but that inference remains cancelable, as demonstrated in (109).

(109) Japanese (Yusuke Yagi, p.c.)

Tatoe anata ga 30-pun mat-tei-ta to shi-temo, zen'in ga
even.if you NOM 30-minute wait-ASP-PST QUOT do-even all NOM
okure-ta wake-de-wa na-i.

be.late-PST conclusion-COP-CNTR NEG-PRS

'Even if you waited for half an hour, that doesn't mean everyone was late. [You just got here too early—everyone else arrived on time.]'

However, the picture changes when *wake de wa nai* is embedded. In such cases, the ISI is no longer cancelable. This is illustrated by the contrast between the standard negation construction in (110-a) and *wake de wa nai* in (110-b): the ISI can be suspended in the former but not the latter.

(110) Japanese (Yusuke Yagi, p.c.)

Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value

plummets to zero.

Bob says:

- a. Chokin-o zenbu ano-kabu-ni tooshi-shi-nak-te yokatta!
 saving-ACC all that-stock-DAT invest-do-NEG-INF glad
 'I'm glad I didn't invest all of my savings in that stock!'
- b. #Chokin-o zenbu ano-kabu-ni tooshi-shi-ta wake-de-wa
 saving-ACC all that-stock-DAT invest-do-PST conclusion-COP-CNTR
 naku-te yokatta!
 NEG-INF glad
 'I'm glad I didn't invest all of my savings in that stock!'

How can we account for the contrast between the cancelable ISI in (109) and the obligatory ISI in (110-b)? I argue that the robustness of the ISI depends on whether a broad-focus interpretation is available. In (109), *wake de wa nai* serves its typical discourse function of rejecting an inference (here, that everyone was late), much like the examples in (107). This suggests that, although the presence of a QP allows for a pragmatically neutral narrow-focus interpretation (108), a broad-focus interpretation remains accessible when the discourse context licenses it. In such cases, the ISI is cancelable because of the broad focus, as predicted by the generalization in (7). By contrast, in (110-b), a broad-focus interpretation appears to be blocked, leaving only the narrow-focus interpretation. As a result, the ISI is non-cancelable, again in line with the generalization in (7).

Two factors might contribute to the unavailability of broad focus in (110-b). First, the felicity conditions for broad focus are not met: the sentence does not reject an inference implied by prior discourse. Nonetheless, *wake de wa nai* can still function as a marker of wide-scope negation with narrow focus on the QP, as shown by the fact that (110-b) is acceptable in contexts where it does not contradict the common ground. Second, the broad-focus interpretation may be inherently incompatible with embedding under emotive factive predicates.³²

In sum, *wake de wa nai* exhibits a mixed behavior. In contexts that support its typical discourse function, it may associate with broad focus, and the resulting ISI is cancelable. However, in standalone sentences or in certain embedded environments, the broad-focus interpretation is pragmatically blocked, effectively requiring narrow focus on the QP. In these cases, the construction behaves like an NFC and gives rise to a presuppositional ISI.

Korean

Korean has two distinct standard negation constructions: the short-form negation employs a pre-verbal negative particle, while in the long-form negation, a negative prefix attaches to an auxiliary, and the lexical verb is followed by the nominalizing suffix *-ci* (known as the suggestive form in traditional grammar). These two constructions are largely interchangeable (Miestamo, 2005, 17).

(111) Korean (Miestamo, 2005, 17)

- a. yong-un tv-lul an pwa-yo.
 Yong-TOP TV-ACC NEG see-POL
 'Yong doesn't watch TV.' (short-form negation)

³² This aligns with an observation made by Bhatt and Homer (2025), who note that Hindi-Urdu *thorī*—which, like *wake de wa nai*, requires a discourse context in which it reacts to a salient assumption or claim—cannot be embedded under predicates such as *be happy* or *be afraid*.

- b. yong-i tv-lul po-ci an-ha-yo.
 yong-NOM TV-ACC see-NMLZ NEG-AUX-POL
 'Yong doesn't watch TV.' (long-form negation)

Besides standard negation, Korean has an NFC, which is superficially similar to the long-form negation, where the contrastive focus marker *-nun* attaches to *-ci*. Propositions of the form 'Not all X are Y' are usually expressed by this NFC (Amiraz, 2025b), as demonstrated in (112).

- (112) Korean (Park and Dubinsky, 2020)
 motun haksayng-i ku mwuncey-lul phwul-ci-nun anh-ss-ta.
 every student-NOM the question-ACC answer-NMLZ-CNTR NEG-PST-DECL
 'Not every student answered the question.'

Example (113-a) involves the long-form standard negation construction, where negation takes wide scope over the preceding object QP.³³ This sentence is felicitous within the given context since it lacks an ISI. By contrast, example (113-b) involves the *V-ci-nun* construction, and it is infelicitous due to the ISI it generates (Eunsun Jou, p.c.).

- (113) Korean (Eunsun Jou, p.c.)
Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value plummets to zero.
 Bob says:
 a. na-uy motun caysan-ul ku cwusik-ey thwucaha-ci ani-ha-ese
 1SG-POSS all saving-ACC that stock-LOC invest-NMLZ NEG-do-CONN
 tahayng-i-ta!
 relief-COP-DECL
 'It's fortunate that I didn't invest all my savings in that stock!'
 (lit. 'I didn't invest all my savings in that stock, so it's a relief.')
- b. #na-uy [motun]_F caysan-ul ku cwusik-ey thwucaha-ci-nun
 1SG-POSS all saving-ACC that stock-LOC invest-NMLZ-CNTR
 ani-ha-ese tahayng-i-ta!
 NEG-do-CONN relief-COP-DECL
 'It's fortunate that I didn't invest ALL my savings in that stock!'

Mandarin Chinese

In Mandarin Chinese, standard negation is expressed using the negative particle *bu* or *meiyou*, which precedes the verb. The choice between the two depends on aspectual distinctions (Miestamo, 2005, 90–91).

- (114) Mandarin Chinese (Miestamo, 2005, 90-91)
 wo bu xihuan tamen.
 I NEG like they
 'I don't like them.'

³³ According to Han et al. (2007), there are two varieties of Korean: one where object universal quantifiers scope below negation, and one where the quantifier takes wide scope. Zeijlstra (2022b) argues that in the latter variety, universal quantifiers are positive polarity items, and they can scope below negation if the entire clause is embedded under an NPI licenser. Thus, negation should be able to take wide scope in (113-a) in both varieties of Korean since *glad* is an NPI licenser (Kadmon and Landman, 1993).

In a distinct negative construction, the negative particle *bu* negates the copula *shi* in clause-initial position. The copula *shi*, which also functions as a focus marker, can associate with two types of foci. It can associate with the immediately following subclausal constituent, as in (115-a), yielding an interpretation similar to an English *it*-cleft. Alternatively, it can associate with the entire clause, as in (115-b), an option that is unavailable for English *it*-clefts (Erlewine, 2022; Jin and Chen, 2022). In the latter case, *bu shi* typically stands in opposition to a subsequent statement (Wang, 2008, 682–683). Thus, *bu shi* roughly spans the functional range of both negated clefts (narrow focus) and *It's not that p* (broad focus) in English.³⁴

(115) Mandarin Chinese

- a. Bu shi [Zhangsan]_F yao lai.
 NEG COP Zhangsan will come
 'It's not Zhangsan who will come.' (Jin and Chen, 2022, 319)
- b. Bu shi [yifu tai shou]_F, shi [ni tai pang le]_F.
 NEG COP clothing too slim COP 2SG too fat PRT
 'It's not that the clothing is too small, it's just that you're too big.'
 (Paul and Whitman, 2008, 424)

Similarly to Japanese *wake de wa nai* (see Appendix: Japanese), the clause-initial *bu shi* construction is pragmatically neutral when used in 'Not all X are Y' sentences (Amiraz, 2025b). For example, (116) is fully natural as a standalone sentence. In such cases, *shi* associates with narrow focus on the QP.

(116) Mandarin Chinese (Fan, 2017, 46)

- Bu shi mei-pi ma dou tiaoguo-le liba.
 NEG COP every-CLF horse all jump-over-ASP fence
 'Not every horse jumped over the fence.'

Because it allows broad focus, this construction is classified as a non-NFC. Like Japanese *wake de wa nai*, however, it shows mixed behavior depending on whether a broad-focus interpretation is available. In unembedded 'Not all X are Y' sentences, *shi* can associate with the entire clause rather than exclusively with the QP. This is illustrated in (117), where the clause negated by *bu shi* contrasts with the subsequent clarifying clause, much like in (115-b). Since this example involves broad focus, the ISI is cancelable, in line with the generalization in (7).

(117) Mandarin Chinese (Danfeng Wu, p.c.)

- Bu shi meige ren dou chidao le, shi ni dao de tai zao le. Qita
 NEG COP every person all be.late PRT COP you arrive DE too early PRT other

³⁴ Although clause-initial *bu shi* often responds to a preceding utterance, it can also appear in discourse-initial position, as shown in (i), where it serves to hedge the speaker's forthcoming statement (Ran, 2013).

(i) Mandarin Chinese (Ran, 2013, 101)

Bu shi wo piping Xiao Wang, ta de biao xian tai rang ren shi wang le.
 NEG COP 1SG criticize Xiao Wang 3SG GEN behavior quite make person disappointed PRT
 'I'm not criticizing Xiao Wang, but his behavior disappointed me so much.'

Additionally, *bu shi* has other uses, including as a negative response particle and a marker of metalinguistic negation (Wang, 2008).

ren dou shi zhunshi dao de.
 person all COP on.time arrive DE
 'It's not that everyone was late, it's that you arrived too early. Everyone else arrived on time.'

By contrast, when used as a standalone sentence (116) or embedded in certain environments, *bu shi* requires narrow focus to be felicitous, since the discourse conditions for broad focus are not satisfied. In such contexts, it effectively behaves like an NFC. This is illustrated in (118-b), where the construction occurs in an embedded clause, and a broad-focus interpretation is unavailable. Narrow focus is therefore required, and the ISI becomes non-cancelable. This contrasts with the floating quantifier construction in (118-a), which is based on standard negation and thus qualifies as a non-NFC. As expected, (118-a) remains felicitous in the same context, since the ISI can be suspended.

(118) Mandarin Chinese (Zeqi Zhao, p.c.)

Context: Mary teaches Intro to Statistics. She accidentally gave the students the exam for Advanced Statistics. She expected that they would all fail, but surprisingly, they all passed.

Mary (to a colleague): I'm really proud of the students for passing the exam. Frankly, ...

- a. wo hen jingya xuesheng-men jingran meiyou quanbu (dou)
 I very surprised student-PL unexpectedly NEG all all
 guake!
 fail
 'I was surprised that the students didn't all fail!'
- b. #wo hen jingya bu-shi suoyou-de xuesheng dou guake-le!
 I very surprised NEG-COP all-DE student all fail-ASP
 'I was surprised that not all the students failed!'

Like in Japanese, it remains an open question whether the broad-focus interpretation in (118-b) is unavailable because it generally resists embedding, or because it requires the support of a subsequent clarifying clause. What is clear, however, is that in this context, *bu shi* requires narrow focus, and the resulting ISI is presuppositional.

Bosnian-Croatian-Serbian

In Bosnian-Croatian-Serbian, standard negation is expressed using the negative particle *ne*, which is placed immediately before the verb (Alexander, 2006, 28).

(119) Bosnian-Croatian-Serbian (Milica Denić, p.c.)

Moji prijatelji ne jedu meso.
 my.NOM.PL.M friend.NOM.PL NEG eat.PRS.3PL meat.ACC
 'My friends don't eat meat.'

Propositions of the form 'Not all X are Y' are typically conveyed using a standard negation construction with VS(O) word order (Amiraz, 2025b), as demonstrated in (120). This construction qualifies as a non-NFC, since it is based on standard negation. When the subject is not a scope-bearing element, this word order pattern is considered pragmatically marked but not specifically associated with narrow focus (Milica Denić, p.c.), further supporting its classification as a non-NFC.

- (120) Bosnian-Croatian-Serbian (Milica Denić, p.c.)
 Ne jedu svi moji prijatelji meso.
 NEG eat.PRS.3PL all.NOM.PL my.NOM.PL.M friend.NOM.PL meat.ACC
 'Not all my friends eat meat.'

Consistent with the generalization in (7), the construction in (121) does not give rise to an ISI and is fully felicitous in the given context.

- (121) Bosnian-Croatian-Serbian (Milica Denić, p.c.)
Context: Mary teaches Advanced Statistics. She asks the students a very basic question from Intro to Statistics, which all of the students should be able to answer at this point. Surprisingly, none of the students knows the answer.
 Mary (to her students):
 Iskreno, čudi me da ne znate svi
 frankly surprise.PRS.3SG 1SG.ACC COMP NEG know.PRS.2PL all.NOM
 odgovor na ovo pitanje!
 answer.ACC to DEM.N.SG.ACC question.ACC
 'Frankly, it surprises me that you don't all know the answer to this question!'

Swedish

In Swedish, standard negation is expressed using the negative particle *inte*, which typically follows the finite verb in main clauses and precedes it in subordinate clauses (Holmes and Hinchliffe, 2013, 569), as shown in (122).

- (122) Swedish (Holmes and Hinchliffe, 2013, 569)
- Per tänker inte laga mat i kväll.
 Per intend.PRS NEG cook food tonight
 'Per isn't planning to cook food tonight.'
 - Per sa att han inte tänkte laga mat i kväll.
 Per say.PST that he NEG intend.PST cook food tonight
 'Per said that he wasn't planning to cook food tonight.'

Propositions of the form 'Not all X are Y' can be conveyed through several constructions (Amiraz, 2025b). Example (123-a) involves inverse scope (a surface-scope reading is either unavailable or strongly dispreferred). Since this construction is based on standard negation, it is classified as a non-NFC. Example (123-b) features a negated QP construction and qualifies as an NFC. Finally, (123-c) is a negated cleft, and is also classified as an NFC.

- (123) Swedish (Alice Bondarenko, p.c.)
- Alla barn gillade inte filmen.
 all.PL children like.PST NEG movie.DEF.SG
 'Not all the children liked the movie.' [* $\forall > \neg$; $\neg > \forall$]
 - Inte alla barn gillade filmen.
 NEG all.PL children like.PST movie.DEF.SG
 'Not all the children liked the movie.'
 - Det var inte alla barn som gillade filmen.
 it be.PST NEG all.PL children who like.PST movie.DEF.SG
 'Not all the children liked the movie.'

Consistent with the generalization in (7), non-NFCs do not trigger obligatory ISIs. For instance, both (124-a) and (124-b), which involve standard negation, are felicitous in the

given context. As discussed in Section 3, (124-c) is structurally ambiguous: the negative particle *inte* may either modify the QP (forming a negated QP construction) or the entire clause (a clause-initial negation construction). In the former case, this construction is an NFC, and the ISI is mandatory, whereas in the latter case, the construction is a non-NFC, making (124-c) felicitous under this interpretation. Lastly, (124-d) is a negated cleft and, as expected for NFCs, triggers an obligatory ISI—unless focus is shifted away from the quantifier to another element of the cleft phrase (see discussion in Subsection 4.2.1).

(124) Swedish (Alice Bondarenko, p.c.)

Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value plummets to zero.

Bob says:

- a. Tack och lov att jag inte investerade alla mina besparingar i
 thank and praise that I NEG invest.PST all.PL my.PL savings in
 den där aktien!
 that there stock
 'Thank goodness I didn't invest all of my savings in that stock!'
- b. Tack och lov att alla mina besparingar inte var
 thank and praise that all.PL my.PL savings NEG be.PST
 investerade i den där aktien!
 invest.PTCP.PL in that there stock
 'Thank goodness all of my savings weren't invested in that stock!'
- c. Tack och lov att inte alla mina besparingar var
 thank and praise that NEG all.PL my.PL savings be.PST
 investerade in den där aktien!
 invest.PTCP.PL in that there stock
 'Thank goodness not all of my savings were invested in that stock!'
- d. #Tack och lov att det inte var [alla]_F mina besparingar som
 thank and praise that it NEG be.PST all.PL my.PL savings which
 var investerade in den där aktien!
 be.PST invest.PTCP.PL in that there stock
 'Thank goodness not all of my savings were invested in that stock!'

Turkish

In Turkish, standard negation is expressed using the negative suffix *-mE*, which attaches to the finite verb, as illustrated in (125-a). For non-verbal predicates, negation is formed using the particle *değil* (Miestamo, 2005, 16), as shown in (125-b).

(125) Turkish (Miestamo, 2005, 16)

- a. gel-me-yecek
 come-NEG-FUT
 '(s)he will not come'
- b. hasta değ-im
 ill NEG-1SG
 'I'm not ill'

Propositions of the form 'Not all X are Y' are typically conveyed using a standard negation construction (Öztürk, 2005; Özyıldız, 2017), which is classified as a non-NFC.

- (126) Turkish (Öztürk, 2005, 131)
 Bütün çocuklar o test-e gir-me-di.
 all children that test-DAT take-NEG-PST
 'Not all of the children took that test.' [* $\forall > \neg$; $\neg > \forall$]

In accordance with the generalization in (7), the example in (127), which features standard negation, is felicitous in the given context and does not give rise to an obligatory ISI.

- (127) Turkish (Ömer Demirok and Deniz Özyıldız, p.c.)
Context: John persuades Bob to invest all of his savings in a stock he claims will be the next Apple. However, before Bob can act on the advice, the stock's value plummets to zero.
 Bob says:
 Ne-y-se ki bütün birikim-im o hisse-de değil!
 what-COP-COND COMP all saving-1SG.POSS that stock-LOC NEG
 'Thank goodness that my savings aren't all invested in that stock!'

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